Principles of online decision-making
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Contents

0.1 Introduction to Online Decision-Making . . . 3
0.1.1 What is online (sequential) decision-making? . . . 3
0.1.2 Three classes of models . . . 3
0.1.3 Probability tools we will need . . . 4
0.1.4 Reinforcement learning vs. supervised learning . . . 5
0.1.5 Key concept: the explore-exploit tradeoff . . . 6
0.1.6 Prerequisites . . . 6
0.2 Optimal Stopping: the Secretary Problem . . . 6
0.2.1 Problem statement . . . 6
0.2.2 Optimal policy: threshold structure . . . 7
0.2.3 General optimal threshold k . . . 8
0.2.4 Random order vs. i.i.d. random variables . . . 9
0.2.5 Does the goal matter? Success probability vs. expected value . . . 9
0.2.6 Explore-exploit tradeoff, revisited . . . 10
0.3 Multi-Armed Bandits . . . 11
0.3.1 A bit of history and motivation . . . 11
0.3.2 Stochastic bandits: definitions, assumptions, vocabulary . . . 12
0.3.3 Unstructured vs. structured bandits . . . 12
0.3.4 Review of some probability notions . . . 14
0.3.5 Back to bandits: regret . . . 15
0.3.6 Objective of the learner: sublinear regret . . . 16
0.3.7 Decomposing the regret . . . 16
0.3.8 Summary . . . 17
0.4 Concentration of Measure . . . 17
0.4.1 Motivation: sums of i.i.d. random variables . . . 17
0.4.2 Large n limits . . . 18
0.4.3 Markov and Chebyshev inequalities . . . 19
0.4.4 Gaussian tail from the CLT . . . 19
0.4.5 Subgaussian random variables . . . 20
0.4.6 Tail of a subgaussian random variable . . . 21
0.4.7 Large deviation bound for finite n . . . 22
0.4.8 Examples of subgaussian random variables . . . 22
0.5 The Explore-then-Commit Algorithm . . . 23
0.5.1 Idea and definitions . . . 23
0.5.2 Simulations: Bernoulli bandit with $k = 2$ . . . 23
0.5.3 Regret bound for ETC with 1-subgaussian rewards . . . 25
0.5.4 Analysis for $k = 2$ . . . 26

0.5.5 Comparison of the bound with simulations . . . 27
0.6 Summary . . . 27
0.7 Recap: Bandits, Regret and Explore-then-Commit . . . 28
0.7.1 Multi-armed bandits . . . 28
0.7.2 Regret . . . 28
0.7.3 Explore-then-Commit (ETC) . . . 28
0.8 Adaptive Explore-then-Commit . . . 29
0.8.1 Key idea: m as a stopping time . . . 29
0.8.2 Confidence intervals: recap . . . 29
0.8.3 Using confidence intervals to decide when to commit . . . 29
0.8.4 Pseudo-code . . . 30
0.8.5 Regret curves of adaptive ETC . . . 30
0.9 Beyond Two Arms: Sequential Elimination . . . 31
0.9.1 ETC for $k > 2$ arms . . . 31
0.9.2 From ETC to sequential elimination . . . 31
0.10 The Upper Confidence Bound (UCB) Algorithm . . . 32
0.10.1 Definition . . . 32
0.10.2 Exploration-exploitation tradeoff in UCB . . . 32
0.10.3 UCB vs. ETC: optimism in the face of uncertainty . . . 33
0.10.4 Regret performance of UCB (simulations) . . . 33
0.11 Regret Bound for UCB . . . 33
0.11.1 Notation: $\hat{\mu}_i(t)$ vs. $\hat{\mu}_{is}$ . . . 34
0.11.2 Proof outline . . . 34
0.11.3 Step 1: the good event G_i . . . 34
0.11.4 Step 2: if G_i holds, then $T_i(n) \leq u_i$ . . . 35
0.11.5 Step 3: G_i^c is unlikely . . . 35
0.11.6 Step 4: choice of u_i and conclusion . . . 36
0.11.7 UCB regret plots . . . 36
0.12 Summary . . . 37

0.1 Introduction to Online Decision-Making

0.1.1 What is online (sequential) decision-making?

Online, or sequential, decision-making refers to settings in which an *agent* repeatedly decides on an action, receives feedback from the environment, and then decides on another action, and so on. Usually, the sequence of decisions coincides with time.

Three ingredients characterize this class of problems:

- **Sequential**: the agent takes a decision, observes feedback, takes another decision, etc.
- **Decision**: a decision is *irrevocable* – once taken, it cannot be undone.
- **Uncertainty**: the environment is not known in advance, so the agent must *probe* it; the future is not known either, so the agent must rely on statistical assumptions.

[Teaching a new class] Consider a class being taught for the first time. The instructor has some prior information (student background, curriculum, goals) and a finite horizon (e.g., 14 weeks of lectures). Uncertainties include the students' background, how content is best acquired, and extraneous effects (other courses, external events). Every lecture reveals new information (questions asked, homework performance, midterm results), based on which the instructor takes actions: tweaking the lectures, and students may drop or join the class.

0.1.2 Three classes of models

This course studies three families of sequential decision problems, in increasing order of complexity.

- **Optimal stopping** (the secretary problem, covered this week): the horizon is finite and the agent must commit to a choice.
- **Bandits** (about 6 weeks): the agent repeatedly decides which *arm* (option) to play; outcomes are random, and there is an optimal arm that is not known a priori. The agent must learn the value of the different options over time.
- **Reinforcement learning** (about 6 weeks): the agent learns to interact with an environment that has a *state* (i.e., memory); outcomes of actions are random and rewards can be delayed.

Example: optimal stopping – finding a parking spot

Consider the problem of finding a parking spot close to a destination, with limited lookahead: if the destination is reached without having found a spot, this is a failure. If a spot is found, the reward equals minus the distance walked to the destination. There is a clear tradeoff: parking too early means a lot of walking, while parking too late risks missing the destination entirely (or arriving too late). How should one park optimally?

Example: multi-armed bandits

In a multi-armed bandit problem, the set of actions is called the set of *arms*. The agent takes a sequence of decisions on which action (arm) to pull, and receives a reward after each action according to some reward function. In a *stochastic* bandit, the reward is random, and is independent of everything except the choice of action taken. The name traces back to Thompson [1933].

[Clinical trial] Consider running a clinical trial without unnecessarily harming patients. The gold standard for clinical trials is to randomize: randomly give each patient the drug or a placebo. There is always a nonzero probability that the medicine does not work, or is even more harmful than the placebo – so when should we decide that this is unlikely enough that we should stop giving placebos?

The classical statistical approach fixes a confidence criterion (a p-value) beforehand, which determines the size n of the trial (how many patients are needed): more patients give higher confidence. But if the medicine works very well (or seems worse than the placebo), many people stay sick needlessly while the trial runs to completion. Ideally, we should be able to abort the trial early once the outcome becomes clear – but what if this apparent trend was only a fluke?

The question is therefore: instead of assigning treatments purely at random, how do we decide, for each new patient arriving sequentially, whether to give the drug or the placebo, using all information available so far?

Remark 1 (The importance of the horizon)**.** Suppose every arm a generates a reward distributed like X_a, independent of everything else. How would you play if $n = 1$? How would you play if $n \to \infty$? This is analogous to portfolio choice: stocks have higher expected return but also higher volatility, while bonds have lower return and lower volatility – investment advice differs for a 20-year-old versus a 50-year-old. The horizon over which the agent needs to perform well is a key parameter of the problem.

Example: reinforcement learning – Atari Pong

Consider learning to play a computer game (e.g., Atari Pong). We do not (or cannot) program the rules of the game and then compute the optimal move analytically. Instead, an algorithm should observe and master the game as quickly as possible by playing and observing. Feedback comes as reward signals (+1 if the agent scores, −1 if the adversary scores). The environment has a *state* (e.g., ball position and velocity, own position), and the right action depends on this state. A key additional difficulty is that signals arrive *with delay*.

0.1.3 Probability tools we will need

Concentration results

Concentration results describe the limiting behavior of probability distributions at different "zoom levels". Suppose $X_1, \ldots, X_n$ is a sequence of i.i.d. random variables, and let $S_n = \sum_{i=1}^{n} X_i$.

- **Law of Large Numbers**: $\dfrac{S_n}{n} \to \mu$ (the average converges to the mean). This is an *asymptotic* statement ($n \to \infty$) and gives no information for a specific finite n.
- **Central Limit Theorem**: $\dfrac{S_n - n\mu}{\sqrt{n}} \to \mathcal{N}(0, \sigma^2)$, characterizing the "typical" (standard) deviation from the mean.
- **Chernoff bound**: $\mathbb{P}\left(\dfrac{S_n}{n} \geq \mu + \epsilon\right) \leq \exp(-f(\epsilon)n)$, giving a more precise tail behavior for a specific finite n, i.e., the probability of very large departures from the mean.

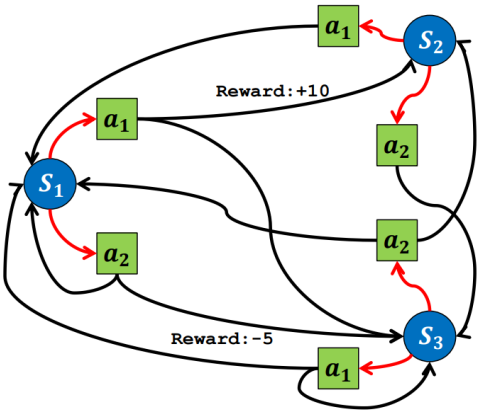
Markov chains

We are familiar with sequences of i.i.d. random variables $X_1, \ldots, X_n$ (LLN, CLT, etc.). To add "a bit of dependence", consider (X_1, X_2, X_3) with $X_1 \perp X_3 \mid X_2$: this defines a *Markov chain*. Now let the number of variables go to infinity, assume homogeneity (the probabilistic relationship between neighboring variables is the same everywhere) and discrete random variables. The chain is then described by a transition matrix P, with $P_{ij} = \mathbb{P}(X_n = j \mid X_{n-1} = i)$.

Depending on P, sets of states can have different long-term behaviors (absorbing, transient, periodic). Under certain conditions on P, the long-term behavior of the chain converges (*ergodicity*); in this ergodic regime, one is interested in the *stationary distribution* π, the normalized solution of the eigenvector problem $\pi = \pi P$.

Markov Decision Processes (MDPs)

An MDP formalizes decision-making in an environment with state. A *policy* π maps a state to an action. The goal is to maximize the reward over a long time horizon.



[Learning to walk] Model: physics of limbs, gravitation, friction with the ground. Actions: move joints. Rewards: getting to some place. Key challenges: large, continuous state and action spaces. This topic will not be covered in detail in this course.

0.1.4 Reinforcement learning vs. supervised learning

In classical supervised learning: labeled data is collected, a model is trained to give good predictions, and at test time the model is deployed and used.

In the real world, data keeps accruing during test/deployment time, which can be used to refine the model; moreover, the system might not be stationary, requiring adaptation (e.g., periodic retraining). Updating the model may also influence the future data distribution: e.g., in a recommender system, people consume the items the system recommends, so future data becomes biased toward recommended items – a potential positive feedback loop.

Reinforcement learning integrates "training" (*explore*) with "testing", i.e., using the model to accomplish something (*exploit*).

0.1.5 Key concept: the explore-exploit tradeoff

Definition 1 (Explore-exploit tradeoff)**. Explore**: collect samples to learn about the environment/model – somewhat like the training set in supervised learning; the agent does not attempt to take good decisions, but to understand the problem.
Exploit: try to take good actions/decisions – somewhat like the test set in supervised learning; the agent does try to take good decisions, even though the model is still noisy.

This tradeoff recurs across all the examples seen so far:

- **Parking**: explore (look for a better spot) vs. exploit (park now).
- **Clinical trial**: explore (give a random treatment) vs. exploit (give the treatment believed best).
- **Pong**: explore (random move) vs. exploit (best-guess move).

This course develops probabilistic models for such situations, algorithms to handle them, and analysis of the optimality of these algorithms.

0.1.6 Prerequisites

This is essentially a probability class; introductory probability is enough, although mathematical maturity helps. It is not a bad idea to take *Modèles stochastiques* in parallel, since there is some overlap (mostly Markov chains) – however, this course (PODM) is self-contained. Solid prerequisites in basic probability are required.

0.2 Optimal Stopping: the Secretary Problem

0.2.1 Problem statement

Definition 2 (Secretary problem, a.k.a. "apartment hunting problem")**.** You are looking for an apartment to rent on a hot market: you must take it or leave it forever, on the spot. You know the number of options in advance: n apartments will be viewed, and you need shelter after the last one. Every apartment has a quality level x_i, drawn i.i.d. from some *unknown* law – equivalently, one may simply assume that the apartments are viewed in a uniformly random order, with no information about the underlying probability law. The only information available at step i is the values $x_1, \ldots, x_i$ seen so far (or, more precisely, their relative ranking).
Decisions are irrevocable: sign the lease immediately, or that apartment is gone forever. The goal is to maximize the probability of selecting the overall best apartment among the n.

This model captures many practical situations with the same flavor, e.g.:

- Resource allocation: how much effort to spend searching and evaluating before hiring a candidate or selecting a vendor.
- Online auctions: updating one's own bid upon observing other bids.
- Acquiring friendships/social contacts over a lifetime.

The problem is also known under other names: the *marriage problem*, the *sultan's dowry*, or the *googol game*.

0.2.2 Optimal policy: threshold structure

Theorem 1 (Form of the optimal policy)**.** *The optimal policy is necessarily of the following form, for some constant threshold k:*

(1) If x_i is not *a new maximum (relative to $x_1, \ldots, x_{i-1}$), continue.*
(2) If x_i is *a new maximum, stop (commit) if $i \geq k$; otherwise continue.*

Justification. **Why (1)?** Stopping on a value that is not a new maximum is guaranteed to fail (the true best has already appeared, or will appear later and we have already committed to a non-optimal value) – this is never a good idea.

Why (2)? We claim the optimal policy continues before k and stops at or after k upon seeing a new maximum, i.e., the policy is *monotonic*: if we decide to stop upon seeing a new maximum-so-far at position i, we must also decide to stop upon seeing a new maximum-so-far at any $j > i$.

To see this, condition on observing the maximum over $x_1, \ldots, x_i$ seen so far (i.e., $x_i > \max(x_1, \ldots, x_{i-1})$). The probability of success if we commit to x_i is exactly i/n: the overall maximum is uniformly distributed over $\{1, \ldots, n\}$, so this conditional probability of success is *increasing* in i. On the other hand, the (unconditional) probability of seeing a new maximum-so-far at step i is *decreasing* in i. Hence, if the tradeoff (value of stopping vs. probability of the opportunity arising) is favorable at i, it is even more favorable at any $j > i$. □

Small cases: $n = 1, 2, 3$

- $n = 1$: nothing to do, commit to the only option.
- $n = 2$: the best is first with probability $1/2$ and second with probability $1/2$. Either commit to the first, or (implicitly) commit to the second (=last) if not; $\mathbb{P}(\text{success}) = 1/2$.
- $n = 3$: the smallest n for which we actually "make a decision". With $k = 2$, we use the comparison between the first and second position to decide whether to take the second or the third apartment.

The table below enumerates all $3! = 6$ orderings of the ranks $\{1, 2, 3\}$ (1 = best) and the resulting success probability for each candidate threshold $k \in \{1, 2, 3\}$.

	$k = 1$	$k = 2$	$k = 3$
	1 2 3	1 2 3	1 2 3
	1 3 2	1 3 2	1 3 2
	2 1 3	2 1 3	2 1 3
	2 3 1	2 3 1	2 3 1
	3 1 2	3 1 2	3 1 2
	3 2 1	3 2 1	3 2 1
$\mathbb{P}(\text{success})$	1/3	1/2	1/3

The threshold $k = 2$ is optimal for $n = 3$, giving $\mathbb{P}(\text{success}) = 1/2$.

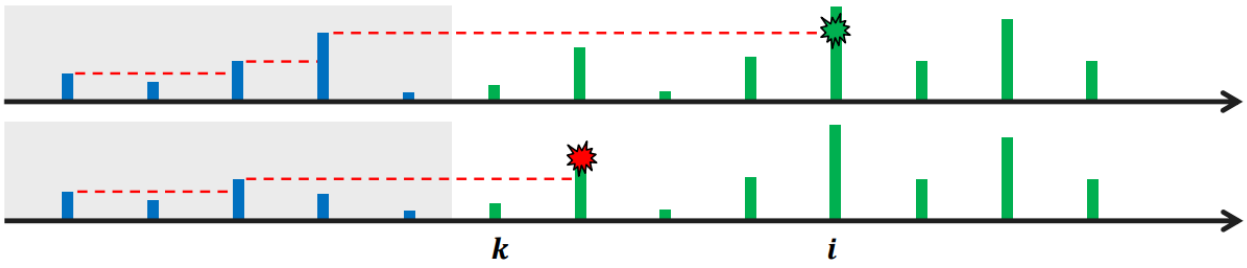
0.2.3 General optimal threshold k

Let U denote the index (position) of the true overall maximum; note that $U \sim \text{Unif}\{1, \dots, n\}$. Let Z denote the output of the policy (its guess for U), and let E be the event "success", i.e., the policy chooses index i and i is indeed the best. Then:

$$\mathbb{P}(E) = \sum_{i=k}^{n} \mathbb{P}(Z = i \cap U = i) = \sum_{i=k}^{n} \mathbb{P}(Z = i \mid U = i)\, \mathbb{P}(U = i), \qquad \mathbb{P}(U = i) = \frac{1}{n}.$$

Computing $\mathbb{P}(Z = i \mid U = i)$. The event $\{Z = i \mid U = i\}$ holds when i is the overall best (so rule (2) is triggered with certainty at step i: success), *provided* the policy has not stopped earlier. This requires that all previous best-so-far values occur *before* k. Conditional on $\{U = i\}$, where is the best value among $x_1, \dots, x_{i-1}$ (not necessarily the second-best overall, which could occur after i – just the best *so far*)? All we know is that it is not at i, so its position is uniformly distributed over $\{1, \dots, i-1\}$. Hence:

$$\mathbb{P}(Z = i \mid U = i) = \mathbb{P}\big(\text{Unif}(i-1) \in \{k-1\}\big) \cdot (k-1) = \frac{k-1}{i-1}.$$



Finite n

Combining the two pieces:

$$\mathbb{P}(E) = \sum_{i=k}^{n} \frac{k-1}{i-1} \cdot \frac{1}{n} = \frac{k-1}{n} \sum_{i=k}^{n} \frac{1}{i-1} = \frac{1}{n}\big(H_{n-1} - H_{k-2}\big),$$

where $H_m = \sum_{j=1}^{m} \frac{1}{j}$ is the m-th harmonic number. Optimizing over k for each n gives:

n	1	2	3	4	5	6	7	8	9	10	20	30	40	50	100
k	1	1	2	2	3	3	3	4	4	4	7	11	15	18	37
$\mathbb{P}(E)$	1	0.5	0.5	0.458	0.433	0.428	0.414	0.410	0.406	0.399	0.384	0.379	0.376	0.374	0.371

Remarkably, $\mathbb{P}(E)$ appears to converge to a positive constant as $n \to \infty$. Also note that neither k nor $\mathbb{P}(E)$ depends on the distribution of the x_i – only on n.

Large n: the $1/e$ rule

Set $x = k/n$. As $n \to \infty$, the sum becomes an integral:

$$\mathbb{P}(E) \xrightarrow{n\to\infty} x \int_{y=x}^{1} \frac{1}{y}\, dy = -x \ln x.$$

Optimizing over x: setting $\dfrac{d\,\mathbb{P}(E)}{dx} = 0$ gives $x = e^{-1} \approx 37\%$.

Theorem 2 (The $1/e$ rule)**.** *As $n \to \infty$, the optimal threshold is $k \approx n/e$, and the optimal success probability converges to $1/e \approx 0.368$. Informally: "look at the first 37% of the candidates without committing, then leap at the next candidate that beats everything seen so far."*

0.2.4 Random order vs. i.i.d. random variables

An important assumption underlying the whole analysis is that we have *no information about future values*. Two ways to phrase this assumption, and why both matter:

- **I.i.d. random variables**: the player does not know the underlying distribution. *Counterexample if this fails*: if we knew the distribution of the x_i, we could directly compute the probability that x_i is the overall maximum; if this probability is already large, there would be no need to wait until k.
- **Rank order**: only the partial order (ranking) of the items seen so far (up to i) is available. *Counterexample if this fails*: if x_i revealed its total rank among all n items, we could simply wait for an item ranked #1 at any time and stop immediately.

The need to "pay" with some samples in order to discover the distribution and correctly judge future samples is what makes the problem interesting, and is baked into these assumptions.

0.2.5 Does the goal matter? Success probability vs. expected value

The optimal policy derived above is specific to the particular criterion adopted, namely maximizing $\mathbb{P}(Z = U)$: it never accepts anything below the maximum-so-far.

Suppose instead we wanted to maximize the *expected value* of the chosen apartment, $\mathbb{E}[x_Z]$ (i.e., rent a *good* apartment, without it being catastrophic if it is not the very best). Does the optimal policy change?

Remark 2. Yes, it does. *Counterexample*: suppose we are close to n and we observe the second-best value seen so far – it is very unlikely that this can still be beaten, so we should take it. In fact, under this new goal, we should accept anything above the *median* of the values seen so far near the end of the sequence. This policy guarantees eventual failure under the previous (probability-of-best) criterion, but performs much better in terms of expected score.

This illustrates a general lesson: the goal (objective function) fundamentally shapes what the optimal policy looks like.

0.2.6 Explore-exploit tradeoff, revisited

The explore-exploit tradeoff introduced earlier is the key underlying concept of this course:

- **Explore**: make potentially bad decisions (with respect to some specified goal) in order to estimate parameters of the system, helping make better decisions in the future.
- **Exploit**: based on these estimates, make good decisions with respect to the specified goal.

In the secretary problem, explore and exploit are *sharply separated*: there are two phases, with an a-priori threshold (k) at which the policy switches from one to the other. In bandit problems and in reinforcement learning, exploration and exploitation will instead meld together more gradually:

- **Bandits**: gradually improve estimates of the rewards of different arms, stop playing some arms earlier than others, and gradually optimize the rewards obtained.
- **Reinforcement learning**: gradually exclude states and actions that are unlikely or costly to lead to rewards, driving the system towards good states.

0.3 Multi-Armed Bandits

0.3.1 A bit of history and motivation

The name "bandit" comes from the slot machine: pulling an arm yields a reward (minus the investment), and $\mathbb{E}[\text{payoff}] < 0$ – hence a "bandit". In the old days, a slot machine used a mechanical random number generator; different machines ("one-armed bandits") can behave subtly differently. The fundamental question is: *which arm to pull*? In particular, if we play for a long time but know nothing at the beginning, how do we use the information gathered along the way optimally?

The field was initiated by William R. Thompson [*Biometrika*, 1933], motivated by clinical trials: sequential decision-making under uncertainty, aiming to minimize unnecessary harm to patients. Bandit models have many applications in online services, where there is a lot of data, a lot of decisions to make, and human intervention is generally not feasible.

Remark 3. The objective of this course is to extract *principles*: clean "vignettes" of often messier real-world situations, chosen for their mathematical tractability. The algorithms studied are not necessarily directly applicable to all real-world situations, but the underlying principles are tools that guide the design of real-world solutions.

Real-world examples

- **A/B-testing web interfaces**: automate design choices for web/online interfaces; reward = user traffic, activity, engagement time, etc.
- **Ad placement**: place ads at the right location and show the right ad at the appropriate time; reward = ad revenue.
- **News recommendations**: serve news stories of interest to every user; reward = clicks, engagement time, etc.
- **Dynamic pricing**: optimize price over time so as to sell the full inventory while maximizing revenue; reward = total revenue.
- **Monte Carlo Tree Search (MCTS)**: used in recent successes of ML approaches to game playing (Atari games, chess, Go, …). A tree captures positions already explored; the algorithm generates a path down the known tree, expands a leaf, plays randomly to the end, and propagates the reward back up. Conceptually, every node contains a bandit algorithm that learns rewards "further down" based on the choice of child node. Over time, higher-reward paths get chosen more often, while uncertainty is still handled explicitly.

Back to the clinical trial example: what might go wrong?

With a purely randomized test, each patient randomly receives the drug or the placebo. At the end of the trial, effectiveness is assessed, e.g., via a hypothesis test: fix a p-value (e.g., 0.05) and a criterion (e.g., probability of recovery twice as high as without treatment); this determines the necessary number of samples. The trial is carried out, then evaluated – success or failure.

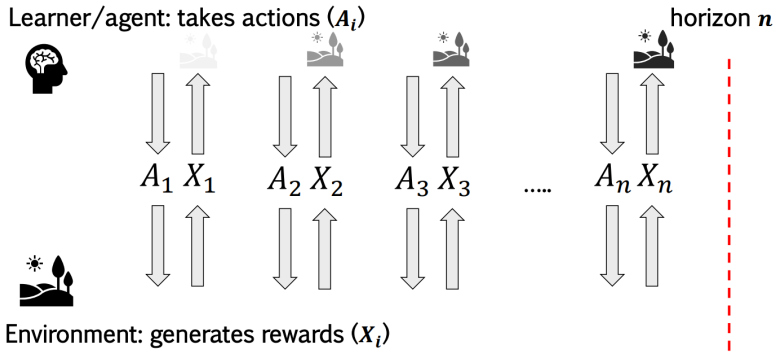
The issue: nothing is done *during* the trial. What if, after only a short time, the data already show a strongly imbalanced outcome between the two arms?

This motivates a *sequential* approach, where decisions can be updated as data comes in, rather than fixed in advance.

0.3.2 Stochastic bandits: definitions, assumptions, vocabulary

The basic interaction loop is as follows: the **learner/agent** takes actions A_i, and the **environment** generates rewards X_i, over a **horizon** n:

$$A_1, X_1, A_2, X_2, A_3, X_3, \ldots, A_n, X_n.$$



Definition 3 (History)**.** The history up to time t is $H_t = (A_1, X_1, A_2, X_2, \ldots, A_t, X_t)$. It contains all the information the learner has so far about the environment.

Definition 4 (Policy)**.** A policy π is a map from history to action: $A_t = \pi(H_{t-1})$. More specifically, the action can be random: π_t is a conditional distribution over A_t given $A_1, X_1, A_2, X_2, \ldots, A_{t-1}, X_{t-1}$ (but it cannot depend on the future).

Definition 5 (Environment)**.** The environment is a map from $A_1, X_1, A_2, X_2, \ldots, A_t$ to X_t (possibly random).

Goal. Maximize the cumulative reward up to the horizon n: $S_n = \sum_{i=1}^{n} X_i$.

Remark 4 (Key assumption)**.** The learner does *not* know the environment at the beginning, although it might know some *class* of possible environments.

Definition 6 (Regret)**.** The regret of the learner relative to a policy π^* is the expected reward under π^* minus the expected reward achieved by the learner – the "cost of ignorance". Usually, π^* is the optimal policy for the environment, i.e., a policy that knows the environment perfectly.

0.3.3 Unstructured vs. structured bandits

Definition 7 (Unstructured bandit)**.** Intuitively, an unstructured bandit is one where pulling one arm provides no information on the reward distribution of other arms. Formally, the environment class is

$$\mathcal{E} = \{\nu = (P_a : a \in \mathcal{A}) : P_a \in \mathcal{M}_a \text{ for all } a \in \mathcal{A}\},$$

where $\mathcal{M}_a$ is a family (set) of possible distributions P_a for action a. In other words, $\mathcal{E} = \bigtimes_{a \in \mathcal{A}} \mathcal{M}_a$ is a *product space*: any combination of elements from the $\mathcal{M}_a$'s is possible. This ensures that knowing anything about P_a provides no information about $P_{a'}$ for $a' \neq a$.

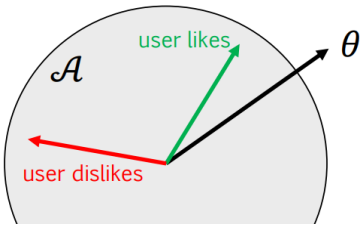
Examples of unstructured bandit classes

Name	Parametric	Definition
Bernoulli	Yes	$\mathcal{B}(\mu_i)$, $\mu_i \in [0,1]$
Uniform	Yes	$\mathcal{U}(a_i, b_i)$, $a_i \leq b_i \in \mathbb{R}$
Gaussian, known variance	Yes	$\mathcal{N}(\mu_i, \sigma^2)$
Gaussian, unknown variance	Yes	$\mathcal{N}(\mu_i, \sigma_i^2)$
Finite variance	No	$\mathbb{V}(P_i) \leq \sigma^2$
Bounded support	No	$\text{Supp}(P_i) \subseteq [a, b]$
Subgaussian	No	P_i is σ-subgaussian

Definition 8 (Structured bandit)**.** Intuitively, a structured bandit has an environment class where the reward of one arm provides information on the reward distribution of other arms.

Examples.

- *"Complementary Bernoulli" bandit*: $\mathcal{A} = \{1, 2\}$, $\mathcal{E} = \{(\mathcal{B}(\theta), \mathcal{B}(1-\theta)) : \theta \in [0,1]\}$. One could learn everything by playing only one arm forever. Example: two mutually exclusive options to click on a webpage.
- *Stochastic linear bandit*: $\mathcal{A} \subset \mathbb{R}^d$ (an uncountably infinite action space), $\theta \in \mathbb{R}^d$, $\mathcal{E} = \{\nu_\theta : \theta \in \mathbb{R}^d\}$ with $\nu_\theta = \{\mathcal{N}(\langle a, \theta \rangle, 1) : a \in \mathcal{A}\}$. Example: a recommender system, where θ captures unknown user preferences and a captures the feature vector of an item; $\langle a, \theta \rangle = a^T\theta$ is the inner product measuring how much the user likes item a.



Thought experiment: the case of no noise

In this idealized setting, we assume that pulling an arm reveals its *exact* expected reward, rather than a noisy sample drawn around it: $r_t = \mu_a$ instead of $r_t = \mu_a + \varepsilon_t$. In other words, a single pull gives perfect information about the arm's true mean, with no randomness to average out. This is of course unrealistic (real feedback is always noisy), but it is a useful thought experiment to understand how much easier the bandit problem becomes once uncertainty is removed.

- **Unstructured bandit**: try each of the k arms once, observe a (deterministic) reward, then play the best arm forever – no more regret is incurred afterwards; the cumulative regret is constant.
- **Structured bandit**: e.g., in the complementary Bernoulli case, there is one permanent winner and one permanent loser, so pulling once suffices to be perfect forever. In the stochastic linear case, getting exact inner products (i.e., projections) requires that the set of actions spans $\mathbb{R}^d$; then d pulls suffice to estimate θ perfectly.

This noiseless setting is not very interesting, nor realistic in most practical settings. **The main topic in bandit theory is dealing with noise in the reward**, which gives rise to a non-zero expected regret "forever".

0.3.4 Review of some probability notions

CDF, CCDF, density

For a continuous random variable A:

$$F_A(a) = \mathbb{P}(A \leq a), \qquad F'_A(a) = \mathbb{P}(A > a) = 1 - F_A(a) \text{ (CCDF)}, \qquad f_A(a) = \frac{dF_A(a)}{da} \geq 0.$$

Independence

Two random variables are independent iff $F_{A,B}(a,b) = F_A(a)F_B(b)$ for all a, b: information on one variable provides no information on the other.

A set of random variables is **pairwise independent** iff every pair is independent. A set of random variables is **mutually independent** iff $F_{A,B,C}(a,b,c) = F_A(a)F_B(b)F_C(c)$ (for three variables; analogously for more): any information on all the other variables provides no information on a target variable.

Remark 5. Mutual independence is *strictly stronger* than pairwise independence!

Let $A, B \sim \mathcal{B}(1/2)$ i.i.d., and let $C = A \oplus B$ (XOR). Contrast the two joint laws below (left: A, B, C constructed this way; right: $A, B, C \sim \mathcal{B}(1/2)$ i.i.d. and mutually independent):

ABC	$\mathbb{P}$
000	1/4
001	0
010	0
011	1/4
100	0
101	1/4
110	1/4
111	0

ABC	$\mathbb{P}$
000	1/8
001	1/8
010	1/8
011	1/8
100	1/8
101	1/8
110	1/8
111	1/8

In the XOR construction, every *pair* (A, B), (A, C), (B, C) is uniform on $\{0,1\}^2$ with all four combinations having probability $1/4$ – i.e., every pair is independent. However, the three variables are *not* mutually independent: knowing any two of them determines the third exactly.

Conditional independence

Definition 9. A and B are conditionally independent given C iff, in the discrete case,

$$\mathbb{P}(A \leq a, B \leq b \mid C = c) = \mathbb{P}(A \leq a \mid C = c)\,\mathbb{P}(B \leq b \mid C = c) \quad \text{for all } a, b, c \ (\mathbb{P}(C = c) > 0),$$

or, in the continuous case, $f_{AB|C}(a, b \mid c) = f_{A|C}(a \mid c)\, f_{B|C}(b \mid c)$ for all a, b, c (with $f_C(c) > 0$).

Intuitively: if C is known, then A provides no further information about B (and vice versa). The notions of pairwise and mutual independence also extend naturally to conditional independence when there are more than two variables.

Expectation

$$\mathbb{E}[X] = \mu_X = \begin{cases} \sum_{x_i} x_i \, \mathbb{P}(X = x_i) & \text{discrete} \\ \int x f_X(x)\, dx & \text{continuous} \end{cases}$$

Some important properties:

- **Linearity**: $\mathbb{E}[A + B] = \mathbb{E}[A] + \mathbb{E}[B]$ for any A, B.
- **Product**: $\mathbb{E}[AB] = \mathbb{E}[A]\,\mathbb{E}[B]$ if $A \perp B$, but this is *not* true in general.
- For $A \geq 0$: $\mathbb{E}[A] = \int_0^\infty F'_A(x)\, dx$ (integral of the CCDF).
- $\mathbb{E}[\mathbf{1}\{E\}] = \mathbb{P}(E)$, where $\mathbf{1}(E)$ is the indicator of event E.

Conditional expectation

Definition 10. $\mathbb{E}[A \mid B = b] = \sum_a a \, \mathbb{P}(A = a \mid B = b) =: f(b)$, and $\mathbb{E}[A \mid B] = f(B)$. Note that this is a *random variable* in B.

Let $B \sim \mathcal{U}(0, 1)$ and $A \sim \mathcal{B}(B)$ (i.e., a Bernoulli whose parameter is itself the random variable B). Then $\mathbb{E}[A \mid B] = B$.

Further properties:

- If $A \perp B$, then $\mathbb{E}[A \mid B] = \mathbb{E}[A]$.
- "Pulling out what is known": $\mathbb{E}[A\, g(B) \mid B] = g(B)\, \mathbb{E}[A \mid B]$, since conditional on $B = b$, $g(B) = g(b)$ is a constant and can be pulled in front of the sum/integral.
- **Tower property (law of total expectation)**: $\mathbb{E}[\mathbb{E}[A \mid B]] = \mathbb{E}[A]$.

[Using the tower property] Let $C \sim \text{Unif}(-1, 1)$, and let $A = \mathcal{N}(C, 1)$, $B = \mathcal{N}(C, 1)$, with $A \perp B \mid C$ (i.e., $A - C - B$ forms a Markov chain). Note that A and B are *dependent* (even though conditionally independent given C). Suppose we need to compute $\mathbb{E}[AB]$. Rather than computing the joint density $f_{AB}(a, b)$ and integrating $\iint ab\, f_{AB}(a, b)\, da\, db$, we can use the tower property:

$$\mathbb{E}[AB] = \mathbb{E}\big[\mathbb{E}[AB \mid C]\big].$$

By conditional independence, $\mathbb{E}[AB \mid C = c] = \mathbb{E}[A \mid C = c]\, \mathbb{E}[B \mid C = c] = c^2$. Hence

$$\mathbb{E}\big[\mathbb{E}[AB \mid C]\big] = \mathbb{E}[C^2] = \int_{-1}^{1} \frac{1}{2} c^2\, dc = \frac{1}{3}.$$

0.3.5 Back to bandits: regret

Definition 11 (Expected and maximum reward)**.** For a stochastic bandit given by $\nu = (P_a : a \in \mathcal{A})$, the expected reward of arm a is

$$\mu_a(\nu) = \int_{-\infty}^{+\infty} x\, dP_a(x) = \mathbb{E}[X], \quad X \sim P_a,$$

and the maximum expected reward is $\mu^*(\nu) = \max_{a \in \mathcal{A}} \mu_a(\nu)$.

Definition 12 (Regret)**.** The regret of policy π on bandit instance ν is

$$R_n(\pi, \nu) = n\mu^* - \mathbb{E}\left[\sum_{t=1}^{n} X_t\right].$$

Note that the regret is *not* a random quantity (the randomness has already been integrated out via the expectation).

Lemma 1. *(a) $R_n(\pi, \nu) \geq 0$ for all policies π.*
(b) The (oracle) policy π choosing $A_t \in \arg\max_a \mu_a$ for all t satisfies $R_n(\pi, \nu) = 0$.
(c) If $R_n(\pi, \nu) = 0$ for some policy π, then $\mathbb{P}(\mu_{A_t} = \mu^*) = 1$ for all $t \in [n]$.

0.3.6 Objective of the learner: sublinear regret

A policy incurring **linear regret** $\Theta(n)$ is trivial to achieve and provides no interesting guarantee – it does not require learning anything, so it is "too easy". A slightly more ambitious objective is **sublinear regret**: we look for policies such that, for all $\nu \in \mathcal{E}$,

$$\lim_{n\to\infty} \frac{R_n(\pi, \nu)}{n} = 0 \qquad \text{(also written } R_n = o(n)\text{)}.$$

This means that, in the long run, the learner chooses the optimal action almost always. (Usually, we are interested in even more ambitious objectives, e.g., controlling the rate at which R_n grows.)

In general, the learner needs to discover the arm (or, for structured bandits, some subset of arms) with the largest mean efficiently. For this, it needs to pull each arm (unstructured case) – or some number of arms (structured case) – a number of times to estimate the environment's parameters. This is exactly the **explore-exploit tradeoff**: exploring (playing all arms to improve the estimators) versus exploiting (playing the best arm(s) to improve the regret).

0.3.7 Decomposing the regret

Definition 13 (Suboptimality gap)**.** The suboptimality gap of action a is $\Delta_a(\nu) = \mu^*(\nu) - \mu_a(\nu)$.

Definition 14 (Action count)**.** The action count of action a by time t is $T_a(t) = \sum_{s=1}^{t} \mathbf{1}\{A_s = a\}$. Note that $T_a(t)$ is a *random variable*, since it depends on the random rewards observed before t (through the policy's choices).

Lemma 2 (Regret decomposition)**.** *For any policy π and environment ν, the regret satisfies*

$$R_n(\pi, \nu) = \sum_{a \in \mathcal{A}} \Delta_a \, \mathbb{E}[T_a(n)].$$

Proof. We rewrite the n-step regret, splitting the sum over time into a sum over arms and using the indicator of which arm is played at each step:

$$R_n = n\mu^* - \mathbb{E}\left[\sum_{t=1}^{n} X_t\right] = \sum_{a \in \mathcal{A}} \sum_{t=1}^{n} \mathbb{E}\big[(\mu^* - X_t)\, \mathbf{1}(A_t = a)\big].$$

By the tower property (conditioning on A_t):

$$= \sum_{a \in \mathcal{A}} \sum_{t=1}^{n} \mathbb{E}\Big[\mathbb{E}\big[(\mu^* - X_t)\, \mathbf{1}(A_t = a) \mid A_t\big]\Big].$$

Now, conditional on A_t, the indicator $\mathbf{1}(A_t = a)$ is a constant and can be pulled out:

$$\mathbb{E}\big[(\mu^* - X_t)\, \mathbf{1}(A_t = a) \mid A_t\big] = \mathbf{1}(A_t = a)\, \mathbb{E}[\mu^* - X_t \mid A_t] = \mathbf{1}(A_t = a)\, (\mu^* - \mu_{A_t}),$$

using the (by-assumption) fact that $\mathbb{E}[X_t \mid A_t] = \mu_{A_t}$. Since the first factor is nonzero only when $A_t = a$, we may replace μ_{A_t} by μ_a:

$$\mathbf{1}(A_t = a)\, (\mu^* - \mu_{A_t}) = \mathbf{1}(A_t = a)\, (\mu^* - \mu_a) = \mathbf{1}(A_t = a)\, \Delta_a.$$

Putting everything together:

$$R_n = \sum_{a \in \mathcal{A}} \sum_{t=1}^{n} \mathbb{E}\big[\mathbf{1}(A_t = a)\big]\, \Delta_a = \sum_{a \in \mathcal{A}} \Delta_a \sum_{t=1}^{n} \mathbb{P}(A_t = a) = \sum_{a \in \mathcal{A}} \Delta_a\, \mathbb{E}[T_a(n)],$$

since $\sum_{t=1}^{n} \mathbf{1}(A_t = a) = T_a(n)$. □

Remark 6 (Intuition)**.** The learner should pull arms with low (or zero) suboptimality gap, and minimize the expected number of pulls of arms with high suboptimality gap. This is not exactly shocking – the challenge of the course is devising algorithms that achieve this efficiently, *without* knowing the gaps Δ_a in advance.

0.3.8 Summary

- The stochastic bandit: a player and an environment interact via actions A_t and (instantaneous) rewards X_t.
- Different assumptions about the possible environments $\mathcal{E}$ are possible (unstructured vs. structured); this assumption is important, because the player needs to figure out everything else by probing, i.e., exploring.
- Regret $R_n(\pi, \nu)$: how much is lost (over horizon n) because the environment is not known in advance.
- Regret decomposition: the regret decomposes as a sum, over all arms, of their suboptimality gap Δ_a times the expected number of times that arm was played, $\mathbb{E}[T_a(n)]$.

Lecture 3 — 2026-09-23: Concentration of Measure and Explore-then-Commit

0.4 Concentration of Measure

0.4.1 Motivation: sums of i.i.d. random variables

To analyze bandit algorithms, we have to deal with sums of random rewards (hence of random regrets). The building block is the following setting.

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables whose mean and variance exist:

$$\mu = \mathbb{E}[X], \qquad \sigma^2 = \mathbb{V}[X] = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big].$$

A natural *estimator* of μ is the empirical mean

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} X_i =: \frac{S_n}{n}.$$

It satisfies:

- $\mathbb{E}[\hat{\mu}] = \mu$: $\hat{\mu}$ is an *unbiased* estimator;
- $\mathbb{V}[\hat{\mu}] = \mathbb{E}\big[(\hat{\mu} - \mu)^2\big] = \dfrac{\sigma^2}{n}$: the variance of the estimator decreases with n.

The question of this chapter is: *how close to μ is $\hat{\mu}$, with what probability, and for a given finite n?* We go through increasingly precise answers: law of large numbers, central limit theorem, Markov/Chebyshev inequalities, and finally exponential (Chernoff-type) bounds valid for finite n.

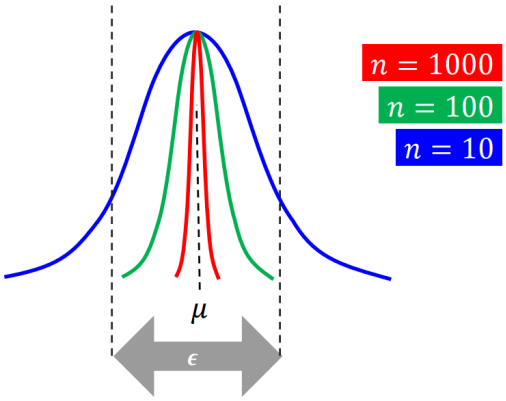
0.4.2 Large n limits

Law of large numbers (LLN)

Theorem 3 (Law of large numbers)**.** *Let $X_1, \dots, X_n$ be i.i.d. with mean μ.*

- ***Weak law:*** *for every $\epsilon > 0$, $\lim_{n\to\infty} \mathbb{P}\big(|S_n - n\mu| \geq n\epsilon\big) = 0$.*
- ***Strong law:*** $\mathbb{P}\left(\lim_{n\to\infty} \frac{S_n}{n} = \mu\right) = 1$.

Remark 7 (Two notions of convergence)**.** The weak law is a statement of *convergence in probability*, the strong law a statement of *almost sure convergence*. The latter is strictly stronger.
Example of a (non i.i.d.) sequence that converges to 0 in probability but not almost surely: let the Y_n be independent with $Y_n = 1$ with probability $1/n$ and $Y_n = 0$ otherwise. Then $\mathbb{P}(Y_n = 1) = 1/n \to 0$, so $Y_n \to 0$ in probability; but since $\sum_n 1/n = \infty$, almost every trajectory ω contains infinitely many 1's, so $Y_n(\omega)$ does not converge to 0.



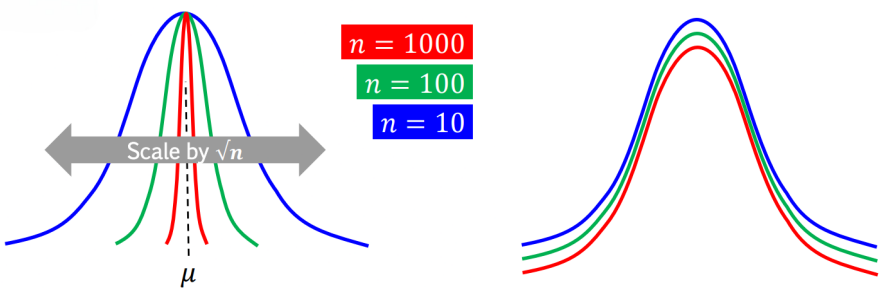
Note that the LLN is an *asymptotic* statement ($n \to \infty$): it says nothing about the probability of a deviation for a specific finite n.

Central limit theorem (CLT)

The LLN says that S_n/n concentrates around μ. The CLT tells us at which *scale*: rescale $\hat{\mu} - \mu$ by the inverse of its standard deviation $(1/\sqrt{n})$, i.e., look at $S_n - n\mu$ divided by $\sqrt{n}$:

$$\frac{S_n - n\mu}{\sqrt{n}} \xrightarrow{n\to\infty} \mathcal{N}(0, \sigma^2).$$

Equivalently, for large n, $\mathbb{P}\big(S_n - n\mu \geq n\epsilon\big) \to \mathbb{P}\big(\mathcal{N}(0,\sigma^2) \geq \epsilon\sqrt{n}\big)$.
This is the primary reason why the Gaussian law is so central (in addition to its stability under addition).



0.4.3 Markov and Chebyshev inequalities

Theorem 4 (Markov's inequality)**.** *For any random variable X and any $\epsilon > 0$,*

$$\mathbb{P}(|X| \geq \epsilon) \leq \frac{\mathbb{E}|X|}{\epsilon}.$$

Proof. Set $Y = |X| \geq 0$ with density f. Then

$$\mathbb{E}[Y] = \int_0^\infty y f(y)\, dy \geq \int_\epsilon^\infty y f(y)\, dy \geq \epsilon \int_\epsilon^\infty f(y)\, dy = \epsilon\, \mathbb{P}(Y \geq \epsilon).$$

□

Theorem 5 (Chebyshev's inequality)**.** *For any random variable X with finite variance and any $\epsilon > 0$,*

$$\mathbb{P}\big(|X - \mathbb{E}X| \geq \epsilon\big) \leq \frac{\mathbb{V}[X]}{\epsilon^2}.$$

Proof. Apply Markov's inequality to the non-negative random variable $(X - \mathbb{E}X)^2$ with threshold ϵ^2. □

Applied to the empirical mean (whose variance is σ^2/n):

$$\mathbb{P}\big(|\hat{\mu} - \mu| \geq \epsilon\big) \leq \frac{\sigma^2}{n\epsilon^2}.$$

- **Pro**: a nice, very general bound: only the mean and the variance are needed, and it holds for every finite n.
- **Con**: the bound is loose, it only decreases as $1/n$.

We need something much stronger for a meaningful analysis of bandit algorithms.

0.4.4 Gaussian tail from the CLT

By the CLT, $\dfrac{S_n - n\mu}{\sqrt{n\sigma^2}} \to \mathcal{N}(0,1)$. What if the normalized sum were *exactly* Gaussian? Let $Z \sim \mathcal{N}(0,1)$; we want to bound

$$\mathbb{P}(Z \geq z) = \int_z^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx.$$

This integral has no closed form, so we look for a bound. Since $x/z \geq 1$ on the domain of integration ($x \geq z > 0$),

$$\int_z^\infty \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx \leq \frac{1}{\sqrt{2\pi}} \int_z^\infty \frac{x}{z}\, e^{-x^2/2}\, dx = \frac{1}{\sqrt{2\pi}\, z} \exp\left(-\frac{z^2}{2}\right).$$

Now, using $\hat{\mu} - \mu = \dfrac{S_n - n\mu}{n}$ and taking $z = \epsilon\sqrt{n/\sigma^2}$, the CLT heuristic gives

$$\mathbb{P}(\hat{\mu} - \mu \geq \epsilon) \;\longrightarrow\; \mathbb{P}\left(Z \geq \epsilon\sqrt{\tfrac{n}{\sigma^2}}\right) \leq \sqrt{\frac{\sigma^2}{2\pi n\epsilon^2}}\, \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right).$$

- **Pro**: much tighter than Chebyshev in general – *exponential* in n versus $1/n$.
- **Con**: this is an *asymptotic* result: the CLT does not allow us to bound the probability for a finite n.

We will need to pay for this limitation by making additional assumptions on the noise: **subgaussianity**.

0.4.5 Subgaussian random variables

Definition 15 (Subgaussian random variable)**.** A random variable X is *σ-subgaussian* if its moment-generating function $M_X(\lambda) = \mathbb{E}[e^{\lambda X}]$ satisfies, for all $\lambda \in \mathbb{R}$,

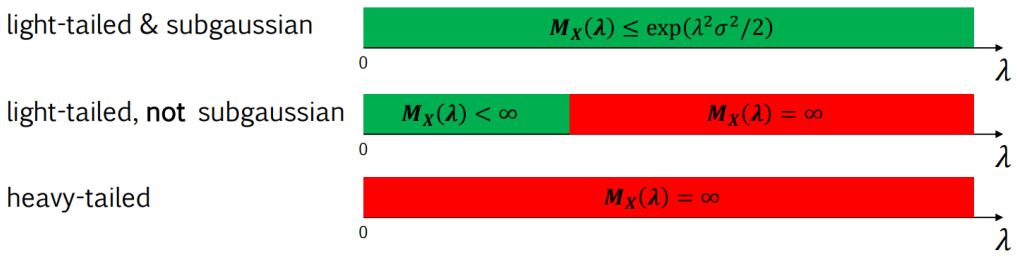
$$M_X(\lambda) = \mathbb{E}\left[e^{\lambda X}\right] \leq \exp\left(\frac{\lambda^2\sigma^2}{2}\right).$$

Equivalently, in terms of the cumulant-generating function,

$$\psi_X(\lambda) = \log M_X(\lambda) \leq \frac{1}{2}\lambda^2\sigma^2$$

for all $\lambda \in \mathbb{R}$.

Remark 8.
- If X is σ-subgaussian, then it is also σ'-subgaussian for any $\sigma' > \sigma$.
- Interpretation: the right-hand side $\exp(\lambda^2\sigma^2/2)$ is exactly the MGF of $\mathcal{N}(0, \sigma^2)$. A subgaussian variable has an MGF dominated by that of a Gaussian, i.e., its tail is at least as light as a Gaussian tail: "nice" noise with "no big excesses".
- Three cases can be distinguished: *heavy-tailed* variables ($M_X(\lambda) = \infty$ for all $\lambda \neq 0$), *light-tailed but not subgaussian* ($M_X(\lambda) < \infty$ on a neighborhood of 0 but $M_X(\lambda) = \infty$ for large $|\lambda|$, or M_X not dominated by the Gaussian one), and *light-tailed and subgaussian* ($M_X(\lambda) \leq \exp(\lambda^2\sigma^2/2)$ for all λ).



Properties of subgaussian random variables

Lemma 3. *Assume X is σ-subgaussian. Then:*
(a) $\mathbb{E}[X] = 0$ and $\mathbb{V}[X] \leq \sigma^2$;
(b) for any $\alpha \in \mathbb{R}$, αX is $|\alpha|\sigma$-subgaussian;
(c) if X_1 is σ_1-subgaussian, X_2 is σ_2-subgaussian and $X_1 \perp X_2$, then $X_1 + X_2$ is $\sqrt{\sigma_1^2 + \sigma_2^2}$-subgaussian.

Remark 9 (Abuse of notation)**.** We sometimes refer to subgaussian random variables that are *not* zero-mean; by this we mean that $X - \mathbb{E}X$ is subgaussian.

Proof of (a), sketch. Compare the Taylor expansions of both sides of $\mathbb{E}[e^{\lambda X}] \leq \exp(\lambda^2\sigma^2/2)$ around $\lambda = 0$. On the left,

$$e^{\lambda X} = 1 + \lambda X + \frac{\lambda^2 X^2}{2} + o(\lambda^2) \implies \mathbb{E}\left[e^{\lambda X}\right] = 1 + \lambda\,\mathbb{E}X + \frac{\lambda^2\,\mathbb{E}X^2}{2} + o(\lambda^2),$$

and, once we know $\mathbb{E}X = 0$ (the linear term in λ must vanish, otherwise the inequality would fail for small λ of the appropriate sign), $\mathbb{E}X^2 = \mathbb{V}X$, so the left side equals $1 + \frac{\lambda^2}{2}\mathbb{V}X + o(\lambda^2)$. On the right,

$\exp(\lambda^2\sigma^2/2) = 1 + \frac{\sigma^2\lambda^2}{2} + o(\lambda^2)$. Hence

$$\mathbb{V}X + \frac{o(\lambda^2)}{\lambda^2/2} \leq \sigma^2 + \frac{o(\lambda^2)}{\lambda^2/2},$$

and taking $\lambda \to 0$ gives $\mathbb{V}X \leq \sigma^2$.
(b) follows since $\mathbb{E}[e^{\lambda\alpha X}] \leq \exp(\lambda^2\alpha^2\sigma^2/2)$; (c) follows from independence: $\mathbb{E}[e^{\lambda(X_1+X_2)}] = \mathbb{E}[e^{\lambda X_1}]\,\mathbb{E}[e^{\lambda X_2}] \leq \exp\big(\lambda^2(\sigma_1^2+\sigma_2^2)/2\big)$. □

0.4.6 Tail of a subgaussian random variable

Theorem 6 (Tail bound)**.** *If X is σ-subgaussian, then for all $\epsilon > 0$,*

$$\mathbb{P}(X \geq \epsilon) \leq \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right).$$

Proof. Let $\lambda > 0$. By monotonicity of exp, then Markov's inequality, then the definition of subgaussianity,

$$\begin{aligned}\mathbb{P}(X \geq \epsilon) &= \mathbb{P}\big(\exp(\lambda X) \geq \exp(\lambda\epsilon)\big)\\ &\leq \mathbb{E}\big[\exp(\lambda X)\big]\,\exp(-\lambda\epsilon)\\ &\leq \exp\left(\frac{\lambda^2\sigma^2}{2} - \lambda\epsilon\right).\end{aligned}$$

This holds for all $\lambda > 0$, so we may choose λ freely: taking $\lambda = \epsilon/\sigma^2$ (the minimizer of the exponent) gives $\exp\big(-\epsilon^2/(2\sigma^2)\big)$. □

The same argument gives the corresponding inequality for the left tail, so that, combining both tails (union bound),

$$\mathbb{P}(|X| \geq \epsilon) \leq 2\exp\left(-\frac{\epsilon^2}{2\sigma^2}\right).$$

Confidence intervals

Inverting this relationship gives *confidence intervals* for X. Fix a small probability (error tolerance) $\delta \in (0,1)$. Solving $\exp(-\epsilon^2/(2\sigma^2)) = \delta$ for ϵ:

$$\mathbb{P}\left(X \geq \sqrt{2\sigma^2\ln(1/\delta)}\right) \leq \delta.$$

For the two-sided version, we use the **union bound** $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$, which yields the factor 2:

$$\mathbb{P}\left(|X| \geq \sqrt{2\sigma^2\ln(2/\delta)}\right) \leq \delta.$$

So, with probability at least $1-\delta$, X concentrates in the confidence interval

$$\left[-\sqrt{2\sigma^2\ln\tfrac{2}{\delta}},\ +\sqrt{2\sigma^2\ln\tfrac{2}{\delta}}\right].$$

We will apply such confidence intervals to characterize the estimators of per-arm reward in different bandit algorithms.

0.4.7 Large deviation bound for finite n

Theorem 7 (Concentration of the empirical mean)**.** *Assume that $X_i - \mu$, $i = 1, \dots, n$, are independent σ-subgaussian random variables. Then for any $\epsilon > 0$,*

$$\mathbb{P}(\hat{\mu} \geq \mu + \epsilon) \leq \exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right).$$

Proof. $\hat{\mu} - \mu = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)$ is $\sigma/\sqrt{n}$-subgaussian (by properties (b) and (c): the sum is $\sqrt{n}\sigma$-subgaussian, and scaling by $1/n$ gives $\sigma/\sqrt{n}$). Apply the tail bound of the previous theorem with parameter $\sigma/\sqrt{n}$. □

Equivalently, $\mathbb{P}(S_n - n\mu \geq n\epsilon) \leq \exp\big(-n\epsilon^2/(2\sigma^2)\big)$. This is a bound similar to what the CLT suggests, but this time it is a true bound valid for every finite n – at the price of applying only to a (much smaller) class of variables. On a semi-log plot, the bound is a straight line in n with slope $-\epsilon^2/(2\sigma^2)$.

Confidence intervals for the mean. Inverting as before:

$$\mathbb{P}\left(\mu \geq \hat{\mu} + \sqrt{\frac{2\sigma^2 \ln(1/\delta)}{n}}\right) \leq \delta, \qquad \mathbb{P}\left(\mu \leq \hat{\mu} - \sqrt{\frac{2\sigma^2 \ln(1/\delta)}{n}}\right) \leq \delta,$$

and, two-sided,

$$\mathbb{P}\left(\mu \in \left[\hat{\mu} - \sqrt{\frac{2\sigma^2 \ln(2/\delta)}{n}},\ \hat{\mu} + \sqrt{\frac{2\sigma^2 \ln(2/\delta)}{n}}\right]\right) \geq 1 - \delta.$$

The width of the interval shrinks as $1/\sqrt{n}$ and grows only as $\sqrt{\ln(1/\delta)}$ when we demand more confidence.

0.4.8 Examples of subgaussian random variables

- **Gaussian is subgaussian**: $X \sim \mathcal{N}(0, \sigma^2)$ has $M_X(\lambda) = \exp(\lambda^2\sigma^2/2)$, hence is σ-subgaussian (with equality in the definition).
- **Bounded is subgaussian**: if $X \in [a, b]$ almost surely, then X is $\frac{b-a}{2}$-subgaussian (Hoeffding's lemma).
  In particular, if $|X| \leq a$ almost surely, then X is a-subgaussian. For instance, a Bernoulli variable is $\frac{1}{2}$-subgaussian.
- **Exponential is *not* subgaussian** (its tail is not light enough!): for $X \sim \mathrm{Exp}(1/\mu)$ (mean μ),

$$M_X(\lambda) = \int_0^\infty \mu^{-1} e^{-x/\mu} e^{\lambda x}\, dx = \begin{cases} (1 - \lambda\mu)^{-1} & \lambda < 1/\mu \\ \infty & \text{otherwise.} \end{cases}$$

  The MGF is infinite for $\lambda \geq 1/\mu$, whereas the subgaussian condition requires it to be finite (and bounded by a Gaussian MGF) for *all* λ.

0.5 The Explore-then-Commit Algorithm

0.5.1 Idea and definitions

Recall the secretary problem: pure exploration for a fixed number of steps, then pure exploitation. We can try a similar policy for bandits: invest some predetermined number of samples to estimate the mean reward of each arm, then play the empirically best arm until the horizon n.

Definition 16 (Estimated reward and number of pulls)**.** For an arm i and a time t,

$$T_i(t) = \sum_{s=1}^{t} \mathbb{I}\{A_s = i\}, \qquad \hat{\mu}_i(t) = \frac{1}{T_i(t)} \sum_{s=1}^{t} \mathbb{I}\{A_s = i\}\, X_s.$$

$T_i(t)$ is the number of pulls of arm i up to time t and $\hat{\mu}_i(t)$ is the estimated (empirical mean) reward of arm i.

Definition 17 (Explore-then-Commit (ETC))**.** Let k be the number of arms and $m \in \mathbb{N}$ a parameter: the number of exploration pulls *per arm* (so mk pulls overall).
Explore-then-Commit(m):
for $t = 1, \dots, n$:

$$A_t = \begin{cases} (t \bmod k) + 1 & \text{if } t \le mk \quad \text{(explore)} \\ \arg\max_i \hat{\mu}_i(mk) & \text{if } t > mk \quad \text{(commit)} \end{cases}$$

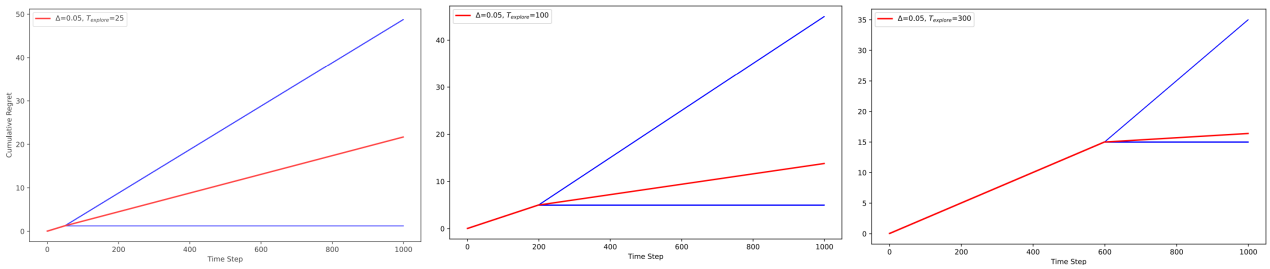
Remark 10.
- The only parameter to set is the number of exploration steps: m per arm, mk overall (the duration of the explore phase).
- The *order* of pulls in the explore phase does not matter, it just needs to be m per arm. One could pull arm 1 m times back-to-back, then arm 2, etc.
- The algorithm itself is generic, but the *analysis* will need to assume subgaussian rewards for all arms.
- After time mk the estimates are frozen, so the arm played from $mk + 1$ to n is the same one at every step (ties broken arbitrarily).

0.5.2 Simulations: Bernoulli bandit with $k = 2$

To build intuition, we look at simulations on a Bernoulli bandit with $k = 2$ arms and suboptimality gap $\Delta = \mu_1 - \mu_2$. In the plots, the **blue** curves are individual trajectories of $\sum_{a \in \mathcal{A}} \Delta_a T_a(t)$, the trajectory-dependent (random) regret, while the **red** curve is $R_n = \sum_{a \in \mathcal{A}} \Delta_a \, \mathbb{E}[T_a(t)]$, the expected regret.

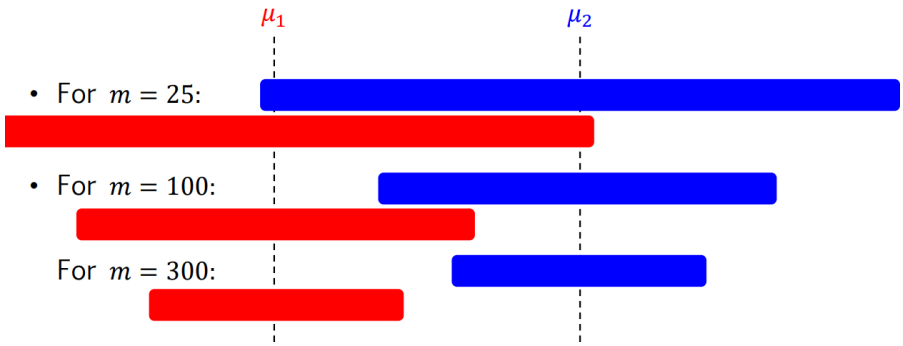
Small gap $\Delta = 0.05$

- **Small** $m = 25$: exploration is cheap but the estimates are noisy, so many trajectories commit to the wrong arm and keep paying linear regret afterwards.
- **Medium** $m = 100$: fewer trajectories fail.
- **Large** $m = 300$: the expected regret is close to its minimum: most trajectories find the right arm and incur no further regret after $t = mk$. The confidence intervals of μ_1 and μ_2 have almost no overlap.



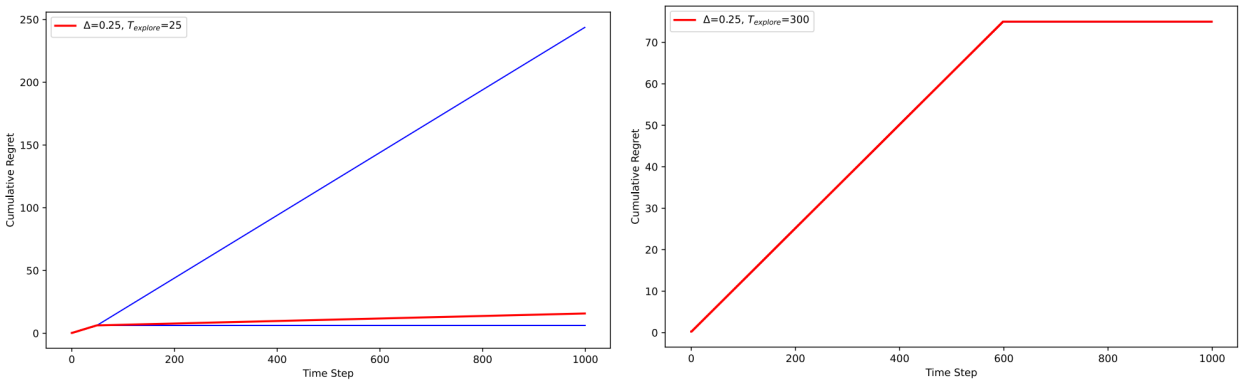
Confidence intervals of the per-arm reward

Recall the confidence interval $\left[\hat{\mu} - \sqrt{\frac{2\sigma^2 \ln(2/\delta)}{m}},\ \hat{\mu} + \sqrt{\frac{2\sigma^2 \ln(2/\delta)}{m}}\right]$ after m pulls. As m grows ($m = 25$, 100, 300), the intervals around μ_1 and μ_2 shrink until they no longer overlap: the algorithm can then tell the arms apart reliably.



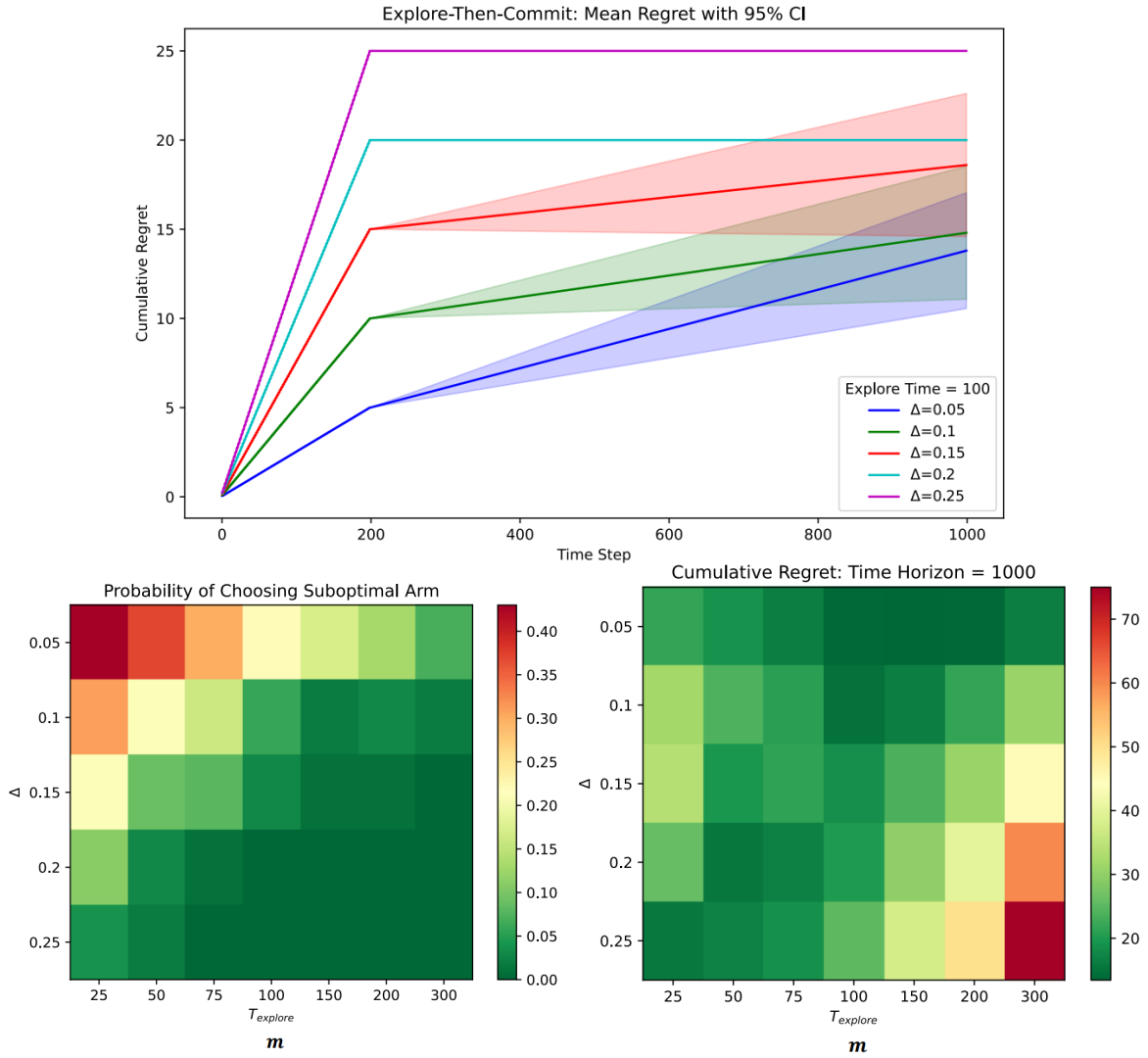
Large gap $\Delta = 0.25$

- **Small** $m = 25$: a larger Δ means the winning arm is more likely to be the better arm, even with few samples.
- **Large** $m = 300$: no more failures in the simulation.



Effect of Δ and m

- **Cumulative regret R_n for different Δ (fixed $m = 100$)**: a larger Δ means more costly exploration (each pull of the bad arm costs Δ), but less risky exploitation (the wrong arm is rarely selected).
- $\mathbb{P}$(**wrong arm at time n**) **as a function of Δ and m**: increasing m helps finding the optimal arm; this is made harder by a smaller Δ.
- **Cumulative regret R_n as a function of Δ and m**: there is a tradeoff. For small Δ one needs a large m; for large Δ a small m suffices. Intermediate Δ gives a higher regret than both the small and the large Δ regimes.



0.5.3 Regret bound for ETC with 1-subgaussian rewards

Theorem 8 (Regret of ETC). *For ETC on a 1-subgaussian bandit (all arms have 1-subgaussian reward noise), for $n \geq mk$,*

$$R_n \leq \underbrace{m \sum_{i=1}^{k} \Delta_i}_{\text{exploration}} + \underbrace{(n - mk) \sum_{i=1}^{k} \Delta_i \exp\left(-\frac{m\Delta_i^2}{4}\right)}_{\text{exploitation}}.$$

Proof. Assume without loss of generality that $\mu_1 = \mu^* = \max_i \mu_i$ (so $\Delta_1 = 0$). By the regret

decomposition lemma (previous lecture),

$$R_n = \sum_{i=1}^{k} \Delta_i \, \mathbb{E}[T_i(n)].$$

Arm i is pulled exactly m times during exploration; afterwards, it is pulled at each of the $n - mk$ remaining steps if and only if it was selected at time $mk + 1$. Hence

$$\begin{aligned}\mathbb{E}[T_i(n)] &= m + (n - mk)\,\mathbb{P}\big(A_{mk+1} = i\big) \\ &\le m + (n - mk)\,\mathbb{P}\Big(\hat{\mu}_i(mk) \ge \max_{j \ne i} \hat{\mu}_j(mk)\Big).\end{aligned}$$

Now, for $i \ne 1$, the event that i beats *all* other arms is contained in the event that it beats arm 1:

$$\begin{aligned}\mathbb{P}\Big(\hat{\mu}_i(mk) \ge \max_{j \ne i} \hat{\mu}_j(mk)\Big) &\le \mathbb{P}\big(\hat{\mu}_i(mk) \ge \hat{\mu}_1(mk)\big) \\ &= \mathbb{P}\Big(\big(\hat{\mu}_i(mk) - \mu_i\big) - \big(\hat{\mu}_1(mk) - \mu_1\big) \ge \Delta_i\Big).\end{aligned}$$

The empirical mean $\hat{\mu}_i(mk)$ is computed from m i.i.d. samples: $m\hat{\mu}_i - m\mu_i$ is a sum of m independent 1-subgaussian variables, hence $m\hat{\mu}_i(mk)$ is $\sqrt{m}$-subgaussian, so $\hat{\mu}_i(mk) - \mu_i$ is $1/\sqrt{m}$-subgaussian. The two arms are sampled independently, so the difference $\big(\hat{\mu}_i - \mu_i\big) - \big(\hat{\mu}_1 - \mu_1\big)$ is $\sqrt{2/m}$-subgaussian. The tail bound then gives

$$\mathbb{P}\Big(\big(\hat{\mu}_i - \mu_i\big) - \big(\hat{\mu}_1 - \mu_1\big) \ge \Delta_i\Big) \le \exp\left(-\frac{\Delta_i^2}{2 \cdot (2/m)}\right) = \exp\left(-\frac{m\Delta_i^2}{4}\right).$$

Plugging everything back into $R_n = \sum_i \Delta_i \, \mathbb{E}[T_i(n)]$ gives the claim. □

Remark 11 (Tradeoff)**.**
- m large ⇒ exploration is costly (first term grows linearly in m);
- m small ⇒ potential to bet on the wrong arm forever (second term is large, it is multiplied by $n - mk$).

This is exactly the explore-exploit tradeoff, now quantified.

0.5.4 Analysis for $k = 2$

Assume again that the first arm is optimal: $\Delta_1 = 0$, and write $\Delta_2 = \Delta$. The bound for general k becomes

$$R_n \le m\Delta + (n - 2m)\Delta \exp\left(-\frac{m\Delta^2}{4}\right) \le m\Delta + n\Delta \exp\left(-\frac{m\Delta^2}{4}\right).$$

Which m minimizes this quantity? Differentiating the right-hand side with respect to m:

$$\Delta - n\Delta \cdot \frac{\Delta^2}{4} \exp\left(-\frac{m\Delta^2}{4}\right) = 0 \iff \exp\left(-\frac{m\Delta^2}{4}\right) = \frac{4}{n\Delta^2},$$

so (ignoring rounding)

$$m = \frac{4}{\Delta^2} \ln\left(\frac{n\Delta^2}{4}\right).$$

The resulting regret bound is of the form

$$R_n \le \Delta + C\sqrt{n},$$

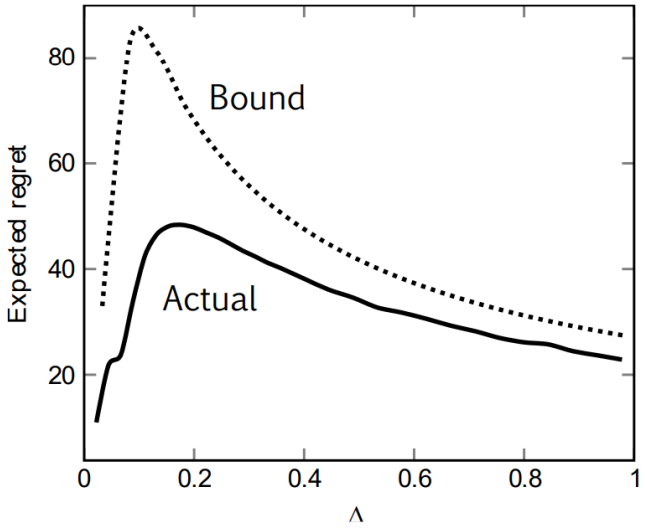
for a universal constant C.

- Without an assumption on Δ, we cannot avoid potentially infinite regret (a single bad pull can cost arbitrarily much). Within a bandit class such that $\Delta \le 1$: $R_n \le 1 + C\sqrt{n}$.

- Replacing Δ by a value result in an example of a **worst-case** (or **problem-independent**) bound: it does not depend on the specific environment instance ν, only on the environment class (here: subgaussian and $\Delta \leq 1$) and on the horizon n.
- **But**: to set the correct m, we do need to know Δ "ahead of time". Can we have a universal algorithm that does well for unknown Δ, i.e., that decides *adaptively* when we have explored enough? (This motivates the algorithms of the next lectures.)

0.5.5 Comparison of the bound with simulations

The bound is relatively loose (due to approximations in the proof) for intermediate suboptimality gaps Δ, but pretty good for larger Δ. It slightly underestimates the critical Δ (the gap at which the actual regret peaks).



0.6 Summary

- **Convergence of random variables**: for sums of i.i.d. variables, the law of large numbers and the central limit theorem describe the asymptotic behavior. Markov and Chebyshev give finite-n bounds, but they are too loose ($1/n$).
- **Tail bound for finite** n: the large deviation (Chernoff-type) bound $\mathbb{P}(\hat{\mu} \geq \mu + \epsilon) \leq \exp(-n\epsilon^2/(2\sigma^2))$, valid for subgaussian noise, and the resulting confidence intervals of width $\sqrt{2\sigma^2 \ln(2/\delta)/n}$.
- **Subgaussian distributions**: "nice" noise, light tail, "no big excesses". It is a strong condition (even the exponential distribution is too heavy). Properties: scaling (αX is $|\alpha|\sigma$-subgaussian) and independent sums (variance parameters add up). Examples: Gaussian, bounded variables.
- **Explore-then-Commit (ETC)**: explore for a fixed window (m pulls per arm), then exploit forever. Tradeoff between expensive exploration and failed exploitation.
- **Upper bound on** R_n **for the** 1**-subgaussian bandit**: $R_n \leq m \sum_i \Delta_i + (n - mk) \sum_i \Delta_i \exp(-m\Delta_i^2/4)$; for $k = 2$ and optimally tuned m, $R_n \leq \Delta + C\sqrt{n}$. Limitation: tuning m requires knowing Δ.

0.7 Recap: Bandits, Regret and Explore-then-Commit

0.7.1 Multi-armed bandits

- **Setup**: k arms, each arm i generates random rewards with distribution P_i and mean μ_i. The best arm has mean μ^*. The reward distributions are *unknown* to the learner at the beginning.
- **Interaction**: at each time $t = 1, 2, \ldots, n$, the learner chooses an action $A_t \in \{1, \ldots, k\}$ and observes a reward $X_t \sim P_{A_t}$. This is a sequential decision-making process.
- **Goal**: achieve *sublinear regret*, i.e., eventually pull the best arm (almost) every time.

0.7.2 Regret

- The **instantaneous regret** at round t is $r_t = \mu^* - \mu_{A_t} = \Delta_{A_t}$. It measures the (expected) suboptimality of the arm played: a bad arm has a large regret, a good arm a small one, and the best arm has zero regret.
- The **cumulative regret** is $R_n = \sum_{t=1}^{n} \mathbb{E}[r_t]$.
- **Regret decomposition**:

$$R_n = \sum_{t=1}^{n} \mathbb{E}[r_t] = \sum_{a \in \mathcal{A}} \Delta_a \, \mathbb{E}[T_a(n)].$$

Remark 12. Sublinear cumulative regret $\Leftrightarrow$ vanishing instantaneous regret. To minimize regret, bad arms must be eliminated quickly, and eventually the best arm(s) must be identified and pulled. But, because of randomness, good and bad arms can *never* be identified with certainty!

0.7.3 Explore-then-Commit (ETC)

- Explore until some fixed time mk (m pulls per arm). The exploration phase gives an estimate $\hat{\mu}_i$ of μ_i for every i.
- If $\hat{\mu}_i \approx \mu_i$, then $i^* = \arg\max_i \hat{\mu}_i$ identifies the best arm. Commit to (keep pulling) i^* for rounds $mk + 1$ to n.

Intuition. If the best arm is correctly identified, there is no regret in the commit phase, hence $R_n \propto m$ (great!). If a suboptimal arm is selected, a constant positive regret is paid at *every* round of the commit phase, hence $R_n \propto n$ (bad!).

Tradeoff. A very small m means $|\hat{\mu}_i - \mu_i|$ is large, so we are likely to choose a suboptimal arm (large regret). A very large m means excessive exploration (also large regret). Tuning m optimally requires knowing the gap Δ in advance.

0.8 Adaptive Explore-then-Commit

0.8.1 Key idea: m as a stopping time

- **Key idea**: think of m as a *stopping time*. It is random and can depend on past data; it signals when to stop exploring and start exploiting. Ideally, we stop when we are sufficiently confident about which arm(s) are optimal.
- With a *fixed* m, two things can go wrong: we might not be sure about the optimal arm at time mk (*under-exploration*), or we might have been very sure about the optimal arm much earlier (*over-exploration*).
- **Idea**: decide when to stop by looking at the data. Keep pulling all arms equally and computing the $\hat{\mu}_i$, until we are (reasonably) certain that one arm is better than all the others.

How do we quantify certainty? With **confidence intervals**.

0.8.2 Confidence intervals: recap

Let $X_1, \dots, X_n$ be i.i.d. samples of a random variable with unknown mean μ, and let $\hat{\mu}_n = \frac{1}{n}\sum_i X_i$ be the estimate of μ. Typically, $\hat{\mu}_n$ is close to μ, but not exactly equal to it. We can say:

$$|\hat{\mu}_n - \mu| \leq \epsilon \quad \text{with probability at least } 1 - \delta,$$

or equivalently, $\mu \in [\hat{\mu}_n - \epsilon, \hat{\mu}_n + \epsilon]$ with high probability.

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Confidence interval ϵ vs. confidence level δ. Suppose the X_i are σ-subgaussian. The subgaussian concentration inequality (lecture 3) states

$$\mathbb{P}\big(|\hat{\mu}_n - \mu| \geq \epsilon\big) \leq 2\exp\left(-\frac{n\epsilon^2}{2\sigma^2}\right) =: \delta.$$

Given ϵ we can compute δ; alternatively, we fix δ and compute ϵ:

$$\epsilon = \sigma\sqrt{\frac{2\ln(2/\delta)}{n}}.$$

- For fixed δ, ϵ decreases as n increases (more data, narrower interval).
- For fixed n, ϵ decreases as δ increases (accepting a larger error probability gives a narrower interval).

Upper and lower confidence bounds. With n the number of samples of the arm considered,

$$\mathrm{UCB}(n,\delta) = \hat{\mu}_n + \sigma\sqrt{\frac{2\log(2/\delta)}{n}}, \qquad \mathrm{LCB}(n,\delta) = \hat{\mu}_n - \sigma\sqrt{\frac{2\log(2/\delta)}{n}}.$$

0.8.3 Using confidence intervals to decide when to commit

How can we confidently decide that one arm is better than another? Pretend that each arm's mean lies in its confidence interval with certainty.

- If the confidence intervals *overlap*, the best arm is undecided: keep exploring.
- If the confidence intervals are *disjoint*, the best arm is clear: commit.

Strategy: keep exploring until the confidence intervals separate.

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Data-dependent exploration time. With sufficient exploration, the confidence intervals eventually separate.

- When $\mu_1 - \mu_2$ is *large*, the intervals separate early, so m tends to be small ($\mathbb{E}[m]$ small).
- When $\mu_1 - \mu_2$ is *small*, the intervals separate only after a long time, so m is large ($\mathbb{E}[m]$ large).

The exploration time is thus data-dependent, not fixed a priori. (Slide example: $m = 20$ is enough to separate the intervals for a large Δ; for the same $m = 20$ the intervals still overlap for a small Δ, and one needs e.g. $m = 100$ to separate them.)

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0.8.4 Pseudo-code

Definition 18 (Adaptive ETC (ETC with confidence intervals))**. Initialize**: $T_a(0) = 0$, $\hat{\mu}_a(0) = 0$ for all arms $a \in \mathcal{A}$; `committed_arm` = `None`.
Hyperparameters: $\delta \in (0, 1)$ and a minimum number of pulls m_0.

For $t = 1, 2, \ldots, n$:

- If `committed_arm` $\neq$ `None`: $A_t =$ `committed_arm`. Else: $A_t = (t \bmod k) + 1$ (round robin).
- **Check for commitment** if $t \geq m_0 k$:
  - compute $\hat{\mu}_a(t)$ for all arms and $a^*(t) = \arg\max_a \hat{\mu}_a(t)$;
  - compute $\mathrm{UCB}_a(t, \delta)$ and $\mathrm{LCB}_a(t, \delta)$ for all arms;
  - if $\mathrm{LCB}_{a^*}(t, \delta) > \mathrm{UCB}_a(t, \delta)$ for all $a \neq a^*$, then `committed_arm` $= a^*$.

Remark 13.
- The empirically best arm is committed to only when its *lower* confidence bound exceeds the *upper* confidence bound of *every* other arm, i.e., when its confidence interval dominates all the others.
- The minimum number of pulls m_0 prevents committing on the basis of (nearly) no data.
- Heuristically, if all the confidence intervals contain the true means (which happens with high probability when δ is small) then $\mu_{a^*} \geq \mathrm{LCB}_{a^*} > \mathrm{UCB}_a \geq \mu_a$ for all $a \neq a^*$, so the commitment is correct.

0.8.5 Regret curves of adaptive ETC

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Salient features of adaptive ETC (simulations on a two-armed bandit):

- the exploration period is *random*;
- the expected cumulative regret is *sublinear*.

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- A larger Δ means the algorithm commits earlier.
- As for fixed ETC, the regret is highest for a *moderate* Δ (e.g. $\Delta = 0.1$ in the simulations).

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Importance of the hyperparameter δ:

- Very small $\delta \Rightarrow$ too conservative (over-exploration).
- Very large $\delta \Rightarrow$ too optimistic: there is a chance of a mistaken commitment (some runs commit to the wrong arm and have linear regret).
- **Rule of thumb**: the larger the horizon n, the smaller δ should be (mistakes are more costly).

0.9 Beyond Two Arms: Sequential Elimination

0.9.1 ETC for $k > 2$ arms

How can the ETC principle be extended to $k > 2$ arms? We need to pull each arm a few times to estimate its reward, so exploration of all arms is necessary. As in the two-armed case, with more exploration the confidence interval of each arm keeps shrinking, eventually the intervals become disjoint, and the best arm becomes clear. There are, however, two different interpretations:

- **View 1** (identify the best arm): keep exploring until the best arm is identified; commit to arm i once its confidence interval dominates the confidence intervals of all other arms.
- **View 2** (eliminate bad arms): keep eliminating arms that are clearly not the best; if the confidence interval of arm i dominates the one of arm j, stop pulling arm j.

These two views are *equivalent with two arms* but *different* in the multi-armed case.

0.9.2 From ETC to sequential elimination

- **Algorithm from View 1** (adaptive ETC): all arms are pulled equally often until the commit time mk. Likely behavior: the worse arms clearly separate out well before the competitive ones, but they keep being pulled $\Rightarrow$ wasteful exploration on bad arms, hence large regret.
- **Algorithm from View 2** (**Sequential Elimination**): initially all arms are pulled in round robin. The algorithm proceeds in *phases*, eliminating one arm per phase; the length of each phase is random. The instantaneous regret decreases from phase to phase, and eventually only the best arm remains.

Sequential elimination is the "right" extension of ETC to multi-armed bandits (to be explored further in the homework).

Comparison on a 4-armed Bernoulli bandit with means $[0.2, 0.4, 0.6, 0.8]$**.**

- *Regret curves*: adaptive ETC shows no commitment in some runs; for sequential elimination the regret slope decreases in phases.
- *Confidence intervals*: in adaptive ETC all the intervals have the same length (all arms are pulled equally), and some overlap may remain; in sequential elimination, the better arms have smaller confidence intervals: they stay in the race longer, are pulled more often, and hence have *narrower* intervals, while eliminated arms keep wide ones.

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0.10 The Upper Confidence Bound (UCB) Algorithm

UCB is a very popular algorithm for multi-armed bandits. Like flexible ETC and sequential elimination it uses confidence bounds, but it leads to better regret bounds in practice.

0.10.1 Definition

Notation. $\hat{\mu}_a(t)$ is the estimate of the mean reward of arm a after t overall pulls and $T_a(t)$ is the number of pulls of arm a at time t ("at time t" refers to after t overall pulls). We assume 1-**subgaussian** rewards.

Definition 19 (Upper confidence bound)**.** The upper confidence bound of arm a at time t is

$$\mathrm{UCB}_a(t,\delta) = \hat{\mu}_a(t) + \sqrt{\frac{2\log(1/\delta)}{T_a(t)}},$$

where $\mathrm{UCB}_a(t,\delta) = +\infty$ initially (when $T_a(t) = 0$).

Definition 20 (UCB algorithm)**.** For $t = 1, 2, \ldots, n$:

- choose $A_t = \arg\max_a \mathrm{UCB}_a(t-1,\delta)$;
- observe X_t and update the $\mathrm{UCB}_a(t,\delta)$.

Remark 14.

- Because $\mathrm{UCB}_a = +\infty$ for an unplayed arm, the algorithm starts by pulling each arm once.
- Only the upper bound is needed (a one-sided bound), hence $\ln(1/\delta)$ rather than $\ln(2/\delta)$ as in the two-sided confidence intervals.
- The only hyperparameter is δ. The theorem below uses $\delta = 1/n^2$.

0.10.2 Exploration-exploitation tradeoff in UCB

The index of an arm decomposes as

$$\mathrm{UCB}_a(t,\delta) = \underbrace{\hat{\mu}_a(t)}_{\text{empirical mean}} + \underbrace{\sqrt{\frac{2\log(1/\delta)}{T_a(t)}}}_{\text{exploration bonus}}.$$

The algorithm pulls an arm more often if it is

- either **promising**: the empirical mean $\hat{\mu}_a(t)$ is large (exploitation), or
- **not well explored**: $T_a(t)$ is small, so the exploration bonus is large (exploration).

Why is this a reasonable strategy?

- With high probability, $\mathrm{UCB}_{a^*}(t,\delta) > \mu_{a^*}$ at all times.
- If a suboptimal arm $a \neq a^*$ is pulled, then $\mathrm{UCB}_a(t,\delta) > \mathrm{UCB}_{a^*}(t,\delta)$.
- With enough pulls, $\mathrm{UCB}_a(t,\delta) \approx \mu_a$ (the bonus vanishes), and $\mu_a < \mu_{a^*} < \mathrm{UCB}_{a^*}(t,\delta)$, so a stops being chosen.
- Hence suboptimal arms are not pulled too many times!

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0.10.3 UCB vs. ETC: optimism in the face of uncertainty

- UCB follows the **Optimism in the Face of Uncertainty (OFU)** principle: it plays assuming that arm rewards are the highest they can *reasonably* be, i.e., it pretends that $\mu_a = \mathrm{UCB}_a(t, \delta)$.
- ETC (and sequential elimination) is more *conservative*: it takes a worst-case approach, where arm rewards could be anything within the respective confidence intervals. It only commits to the best arm if it is "sure" that it is the best (its interval must dominate all the others), and only eliminates a bad arm if it is "sure" it is not the best.
- UCB has a **soft** transition from exploration to exploitation.
- ETC has a **hard** transition (explore phase vs. commit phase), which makes it difficult to correct a mistake.

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0.10.4 Regret performance of UCB (simulations)

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Points to note:

- For each sample path, the regret growth decays gently with time (contrast with ETC, where the slope drops abruptly at commitment, or never drops if the wrong arm is chosen).
- Exploration never stops completely.
- The mean regret is sublinear. Is it $O(\sqrt{n})$? $O(\log n)$? (The theorem below answers: $O(\log n)$ in the instance-dependent sense.)

0.11 Regret Bound for UCB

Theorem 9 (Regret of UCB)**.** *The UCB algorithm applied to a stochastic* 1*-subgaussian bandit with* k *arms and an error tolerance* $\delta = 1/n^2$ *satisfies*

$$R_n \leq 3 \sum_{i=1}^{k} \Delta_i + \sum_{i:\, \Delta_i > 0} \frac{16 \ln(n)}{\Delta_i}.$$

Remark 15.

- The first term, $3 \sum_i \Delta_i$, is a constant independent of n; the second term is *logarithmic* in n. Hence UCB achieves **logarithmic** (and in particular sublinear) regret.
- The bound is *instance-dependent*: it depends on the gaps Δ_i. A smaller gap makes an arm harder to distinguish from the best arm and increases the regret bound ($\propto 1/\Delta_i$). Note that this bound alone is not informative when some $\Delta_i \to 0$.
- *(Side remark, not on the slides.)* Since the regret of an arm with gap $\Delta_i < \Delta_0$ is at most $n\Delta_0$ in total, splitting the arms according to their gap and optimizing Δ_0 gives a worst-case (problem-independent) bound of order $O(\sqrt{kn \ln n})$.
- Compared with ETC, UCB needs to know neither Δ nor a tuned m.

0.11.1 Notation: $\hat{\mu}_i(t)$ vs. $\hat{\mu}_{is}$

- $\hat{\mu}_i(t)$ is the estimator of arm i *at time* t.
- $\hat{\mu}_{is}$ is the estimator *after* s *pulls of arm* i, i.e., the average of s i.i.d. samples (the first s rewards generated by arm i).
- The two are related by $\hat{\mu}_i(t) = \hat{\mu}_{iT_i(t)}$. Remember that $T_i(t) \leq t$ for all t.

The key point is that $\hat{\mu}_{is}$ for a *fixed, deterministic* s is an average of s i.i.d. samples, so the concentration results of lecture 3 apply to it directly, whereas $\hat{\mu}_i(t)$ involves the random number of pulls $T_i(t)$.

Inserer ici slide 25

0.11.2 Proof outline

Inserer ici slide 24

Without loss of generality, assume that arm 1 is optimal ($\mu_1 = \mu^*$) and fix a suboptimal arm i ($\Delta_i > 0$). By the regret decomposition, it suffices to bound $\mathbb{E}[T_i(n)]$. The proof has four steps:

1. Define a "good event" G_i such that: (1) μ_1 is *always* (for all $t \in [n]$) below its own UCB; and (2) μ_1 is above the UCB of arm i after u_i pulls of arm i.
2. Show that if G_i holds, then $T_i(n) \leq u_i$.
3. Show that if u_i is large enough, then G_i is very likely.
4. Make a specific choice for u_i that allows us to bound $\mathbb{E}[T_i(n)]$ (without conditioning on G_i).

0.11.3 Step 1: the good event G_i

Let $u_i \in \mathbb{N}$ be a (deterministic) number to be chosen later, and define

$$G_i = \left\{ \mu_1 < \min_{t \in [n]} \mathrm{UCB}_1(t, \delta) \right\} \cap \left\{ \hat{\mu}_{iu_i} + \sqrt{\frac{2}{u_i} \ln \frac{1}{\delta}} < \mu_1 \right\}.$$

- First event: μ_1 is always inside the confidence interval of arm 1 (more precisely, below its UCB).
- Second event: at a time t where $T_i(t) = u_i$, the UCB of arm i does not exceed μ_1.

Decomposing the expectation according to G_i and using $T_i(n) \leq n$ on G_i^c,

$$\mathbb{E}[T_i(n)] = \mathbb{E}\big[\mathbb{I}\{G_i\}\, T_i(n)\big] + \mathbb{E}\big[\mathbb{I}\{G_i^c\}\, T_i(n)\big] \leq u_i + \mathbb{P}(G_i^c)\, n.$$

There is a tradeoff in the choice of u_i:

- u_i too large $\Rightarrow$ the regret bound is too large;
- u_i too small $\Rightarrow \mathbb{P}(G_i^c)$ is too large $\Rightarrow$ the regret bound is too large.

Inserer ici slide 26

0.11.4 Step 2: if G_i holds, then $T_i(n) \le u_i$

Proof. By contradiction: assume that G_i holds and $T_i(n) > u_i$. Then arm i was pulled more than u_i times before n, so there is a round t at which arm i was pulled for the (u_i+1)-th time, i.e., $T_i(t-1) = u_i$ and $A_t = i$. At this round,

$$\begin{aligned}
\mathrm{UCB}_i(t-1,\delta) &= \hat{\mu}_i(t-1) + \sqrt{\frac{2\ln(1/\delta)}{T_i(t-1)}} \\
&= \hat{\mu}_{iu_i} + \sqrt{\frac{2}{u_i}\ln\frac{1}{\delta}} \qquad (T_i(t-1) = u_i) \\
&< \mu_1 \qquad (\text{second event of } G_i) \\
&< \mathrm{UCB}_1(t-1,\delta) \qquad (\text{first event of } G_i).
\end{aligned}$$

Hence arm 1 has a strictly larger UCB than arm i at time $t-1$, so the algorithm could not have chosen arm i at time t: contradiction! □

Inserer ici slide 27

0.11.5 Step 3: G_i^c is unlikely

By De Morgan,

$$G_i^c = \left\{\mu_1 \ge \min_{t\in[n]} \mathrm{UCB}_1(t,\delta)\right\} \cup \left\{\hat{\mu}_{iu_i} + \sqrt{\frac{2}{u_i}\ln\frac{1}{\delta}} \ge \mu_1\right\}.$$

We bound the probability of each of the two events.

First event. Since $\mathrm{UCB}_1(t,\delta) = \hat{\mu}_{1T_1(t)} + \sqrt{2\ln(1/\delta)/T_1(t)}$ and $T_1(t) \in [n]$, the minimum over the n rounds $t \in [n]$ is at least the minimum over all possible numbers of pulls $s \in [n]$ (the latter ranges over at least as many values, which can only lower the minimum):

$$\left\{\mu_1 \ge \min_{t\in[n]} \mathrm{UCB}_1(t,\delta)\right\} \subset \left\{\mu_1 \ge \min_{s\in[n]} \left(\hat{\mu}_{1s} + \sqrt{\frac{2}{s}\ln\frac{1}{\delta}}\right)\right\} = \bigcup_{s\in[n]} \left\{\mu_1 \ge \hat{\mu}_{1s} + \sqrt{\frac{2}{s}\ln\frac{1}{\delta}}\right\}.$$

Each $\hat{\mu}_{1s}$ is an average of s i.i.d. samples, so by the union bound and the subgaussian concentration inequality (with $\epsilon = \sqrt{2\ln(1/\delta)/s}$, so that $\exp(-s\epsilon^2/2) = \delta$),

$$\mathbb{P}\left(\mu_1 \ge \min_{t\in[n]} \mathrm{UCB}_1(t,\delta)\right) \le \sum_{s=1}^{n} \mathbb{P}\left(\mu_1 \ge \hat{\mu}_{1s} + \sqrt{\frac{2}{s}\ln\frac{1}{\delta}}\right) \le n\delta.$$

Second event. Here we need an assumption on u_i: assume that u_i is large enough so that

$$\Delta_i - \sqrt{\frac{2\ln(1/\delta)}{u_i}} \ge c\Delta_i \qquad \text{for some constant } 0 < c < 1.$$

Using $\mu_1 = \mu_i + \Delta_i$,

$$\begin{aligned}
\mathbb{P}\left(\hat{\mu}_{iu_i} + \sqrt{\frac{2}{u_i}\ln\frac{1}{\delta}} \ge \mu_1\right) &= \mathbb{P}\left(\hat{\mu}_{iu_i} - \mu_i \ge \Delta_i - \sqrt{\frac{2}{u_i}\ln\frac{1}{\delta}}\right) \\
&\le \mathbb{P}\big(\hat{\mu}_{iu_i} - \mu_i \ge c\Delta_i\big) \\
&\le \exp\left(-\frac{u_i c^2 \Delta_i^2}{2}\right),
\end{aligned}$$

where the last step is the Chernoff bound for subgaussian averages (u_i is the number of terms in $\hat{\mu}_{iu_i}$).

Combining the two bounds (union bound):

$$\mathbb{P}(G_i^c) \le n\delta + \exp\left(-\frac{u_i c^2 \Delta_i^2}{2}\right).$$

Inserer ici slide 28

Inserer ici slide 29

0.11.6 Step 4: choice of u_i and conclusion

Back to the bound on the expected number of pulls:

$$\mathbb{E}[T_i(n)] \le u_i + \mathbb{P}(G_i^c)\, n \le u_i + n\left(n\delta + \exp\left(-\frac{u_i c^2 \Delta_i^2}{2}\right)\right).$$

We set u_i as small as possible given the constraint $\Delta_i - \sqrt{2\ln(1/\delta)/u_i} \ge c\Delta_i$, i.e.,

$$u_i = \left\lceil \frac{2\ln(1/\delta)}{(1-c)^2\Delta_i^2} \right\rceil.$$

(There is no guarantee that this is $\le n$, but if it is not, the bound on $\mathbb{E}[T_i(n)]$ holds trivially since $T_i(n) \le n$.)
Recall $\delta = 1/n^2$, so $\ln(1/\delta) = 2\ln n$ and $n^2\delta = 1$. The exponential term becomes

$$\exp\left(-\frac{u_i c^2 \Delta_i^2}{2}\right) \le \exp\left(-\frac{c^2 \ln(1/\delta)}{(1-c)^2}\right) = \exp\left(-\frac{2c^2 \ln n}{(1-c)^2}\right) = n^{-\frac{2c^2}{(1-c)^2}}.$$

Therefore

$$\mathbb{E}[T_i(n)] \le \frac{2\ln(1/\delta)}{(1-c)^2\Delta_i^2} + 1 + 1 + n^{1-\frac{2c^2}{(1-c)^2}},$$

where one "+1" comes from the ceiling in u_i and the other from $n^2\delta = 1$. We choose c such that the last term does not blow up, e.g., $c = 0.5$: then $\frac{2c^2}{(1-c)^2} = 2$, so $n^{1-2} = 1/n \le 1$ and $\frac{2\ln(1/\delta)}{(1-c)^2} = \frac{4\ln n}{1/4} = 16\ln n$. Hence

$$\mathbb{E}[T_i(n)] \le 3 + \frac{16\ln n}{\Delta_i^2}.$$

Finally, by the regret decomposition,

$$R_n = \sum_{i:\,\Delta_i>0} \Delta_i\, \mathbb{E}[T_i(n)] \le 3\sum_{i=1}^{k} \Delta_i + \sum_{i:\,\Delta_i>0} \frac{16\ln n}{\Delta_i}. \qquad \blacksquare$$

Inserer ici slide 30

0.11.7 UCB regret plots

Simulations with Bernoulli rewards of means $[0.5 - \Delta/2,\ 0.5 + \Delta/2]$:

- The regret is sublinear for all values of Δ, given a fixed hyperparameter δ.
- The cumulative regret increases inversely with Δ (a smaller gap is harder).
- The slope gradually reduces with time, signifying the shift from exploration to exploitation.
- The initial regret slope is higher for a large Δ, due to the more expensive exploration (each pull of the bad arm costs Δ).

Inserer ici slide 31

UCB vs. ETC (figure from Chapter 7 of Lattimore & Szepesvári): UCB performs similarly to ETC, *without knowing the optimal* m (i.e., without knowing the suboptimality gap).

Inserer ici slide 32

0.12 Summary

- **Explore-then-Commit with a flexible stopping time**: the exploration phase is chosen adaptively, based on data (commit when the confidence interval of the empirically best arm dominates all the others); it gives sublinear regret, and the hyperparameter δ trades off over-exploration (too small) against mistaken commitment (too large). It extends to **sequential elimination** for multiple arms.
- **Main takeaway: the UCB algorithm**:
  - quantifies uncertainty with confidence intervals: $\mathrm{UCB}_a(t,\delta) = \hat{\mu}_a(t) + \sqrt{2\log(1/\delta)/T_a(t)}$;
  - acts *optimistically in the face of uncertainty* (always pulls the arm with the largest UCB);
  - smooth transition from exploration to exploitation (no hard commitment, no knowledge of Δ needed);
  - regret bound (with $\delta = 1/n^2$, 1-subgaussian rewards): $R_n \le 3\sum_i \Delta_i + \sum_{i:\Delta_i>0} 16\ln(n)/\Delta_i$.
- Proof technique: good event G_i (the optimal arm's mean always below its UCB, and arm i's UCB below μ_1 after u_i pulls) $\Rightarrow T_i(n) \le u_i$; bound $\mathbb{P}(G_i^c)$ by a union bound and Chernoff; choose $u_i \propto \ln(n)/\Delta_i^2$.
- Reading: Lattimore & Szepesvári, Chapter 7 (especially Theorem 7.1).